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Effectiveness of Monetary Policy Transmission in Morocco amid Reforms to the Monetary Policy Framework

Prepared by Nadia Mounir and Hannah Brown

WP/26/223

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IMF Working Paper
Middle East and Central Asia Department

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Prepared by Nadia Mounir and Hannah Brown

Authorized for distribution by Laura Jaramillo Mayor

October 2026

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ABSTRACT: This paper assesses the effectiveness of Morocco's monetary policy transmission since 2018 in light of reforms aimed at modernizing the policy framework. Over the past decade, Morocco has introduced greater exchange rate flexibility, strengthened the legal and operational focus on price stability, and enhanced central bank communication—measures intended to anchor inflation expectations and improve transmission. Drawing on several years of post-reform data, we use a two-step empirical strategy: (i) identifying exogenous monetary policy shocks through a Taylor-rule-based approach leveraging Bank Al-Maghrib's forecasts, and (ii) estimating their dynamic effects via local projections, a flexible approach well suited to settings with structural change and limited data. Our findings indicate that transmission to output and inflation has strengthened post-reform: a 1 percentage point policy rate hike reduces output by about 0.3 percent and inflation by 0.1 percent, compared to 0.1 and 0.05 percent pre-reform, respectively. However, responses remain delayed and muted relative to benchmark estimates for other emerging markets. Analysis of transmission channels shows the interest rate channel is most active, the credit channel is partially effective, and the exchange rate channel is stronger post-reform, while the asset price and expectations channels remain weak. These findings provide important insights for Morocco's ongoing transition to an inflation targeting regime.

RECOMMENDED CITATION: Mounir, Nadia, and Hannah Brown. 2026. "Effectiveness of Monetary Policy Transmission in Morocco amid Reforms to the Monetary Policy Framework." IMF Working Paper No. 26/223

JEL Classification Numbers:	E52, E31, E44, F31, F41
Keywords:	Monetary policy transmission; inflation targeting; exchange rate flexibility; transmission channels; monetary policy framework; Morocco.
Authors' email addresses:	nmounir@IMF.org; hbrown@IMF.org

WORKING PAPERS

Effectiveness of Monetary Policy Transmission in Morocco amid Reforms to the Monetary Policy Framework

Prepared by Nadia Mounir and Hannah Brown[1]

[1] The authors would like to thank Laura Jaramillo for her guidance and support, as well as participants in the IMF MCD Insights Seminar series for their helpful comments and suggestions. The authors are also grateful to staff at Bank Al-Maghrib for their valuable feedback and engaging discussions.

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1. Introduction

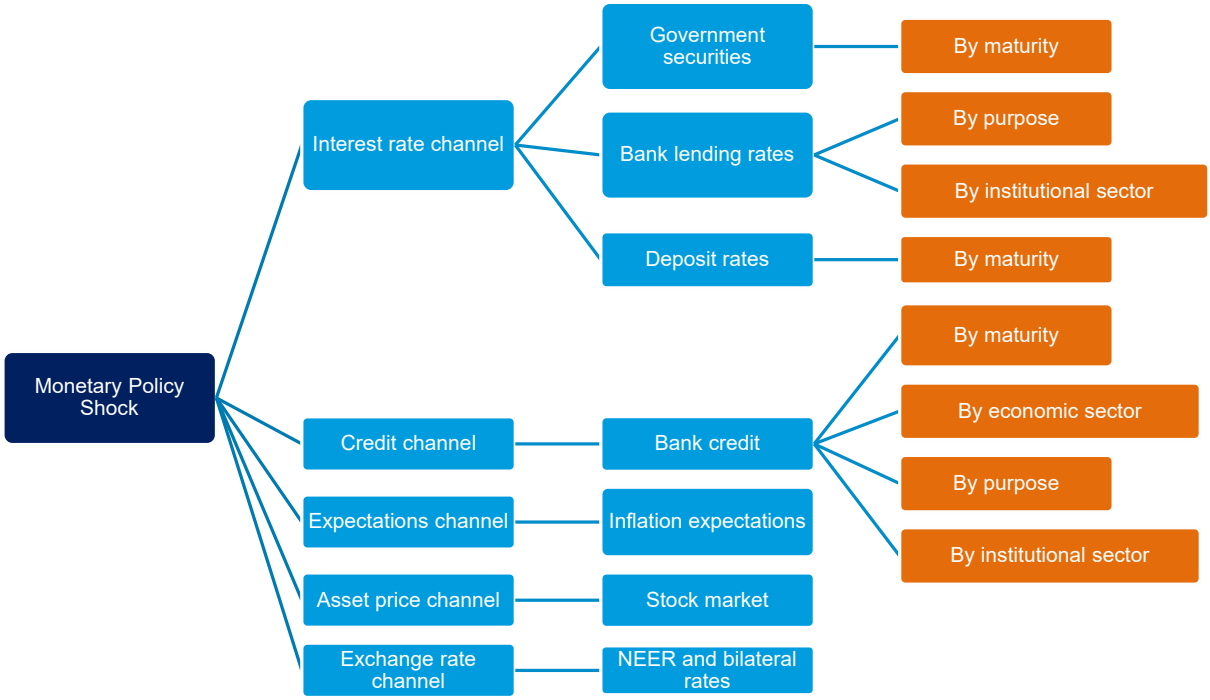
Understanding the monetary policy transmission mechanism is essential for assessing the effectiveness of monetary policy tools in achieving price stability and supporting sustainable economic growth. In Morocco, this question has gained renewed importance as the country is undertaking significant reforms to modernize its monetary policy framework and progress in the transition to an inflation targeting regime (IT). Over the past decade, Morocco has implemented a series of structural and operational reforms that mark a decisive shift toward this objective. These include greater exchange rate flexibility—through successive widening of the dirham's fluctuation band in 2018 (+/- 2.5 percent) and 2020 (+/- 5 percent)—alongside a strengthened legal and operational focus on price stability as a primary objective, and enhanced communication and transparency. Together, these measures aim to anchor inflation expectations, clarify the central bank's reaction function, and improve the transmission mechanism, with greater exchange rate flexibility enhancing the exchange rate channel and the economy's responsiveness to external shocks and relative price signals.

These reforms represent critical milestones and raise important questions about how the transmission mechanism has evolved. This paper assesses the effectiveness of Morocco's monetary policy transmission in the pre- and post-reform periods by leveraging several years of post-reform data and applying a unified empirical framework that jointly assesses macroeconomic effects and key transmission channels—a novel contribution relative to most previous studies for Morocco, which examined individuals channels in isolation and pre-dated recent reforms, and therefore do not capture post-reform transmission dynamics.

The analysis examines key transmission mechanisms, each mapped into a broad set of financial indicators used in the empirical assessment, as summarized in Figure 1.1. Table 1.1 summarizes key empirical studies on monetary policy transmission in Morocco, providing details on sample periods, channels analyzed, methodologies, and main findings. These studies consistently show that, prior to recent reforms, transmission was incomplete and slow, with weak effects on output and inflation (e.g. Bennouna, Lahlou and Mossadak, 2016; Bennouna and Bounader, 2018; Bennouna, 2018). While the interest rate and credit channels were relatively more active, pass-through to lending and market rates remained partial, and the exchange rate channel was largely constrained by the fixed regime (e.g. Queyranne, Baska, Bazines and Abdulkarm, 2021). Cross-country evidence (e.g. Brandão-Marques et al., 2020; Deb et al., 2023; IMF, 2023; Checo, Grigoli, and Sandri, 2024) reinforces these findings, showing that monetary policy transmission in emerging markets can be broadly comparable to that in advanced economies when shocks are properly identified and supported by inflation-targeting frameworks. These studies highlight that the exchange rate and expectations channels play a more prominent role under flexible exchange rate regime. They also show that, while institutional quality, central bank credibility and financial development amplify transmission effectiveness, the monetary policy framework itself is a key determinant of macroeconomic impact.[2] This evidence underscores the need for a post-reform assessment that captures the evolving dynamics of transmission channels in Morocco's modernized monetary policy framework.

[2] Brandão-Marques et al., 2020 show that monetary policy frameworks matter more than other structural country characteristics for the transmission of interest rate shocks to output. Responses tend to be stronger in countries that adopted an inflation targeting regime, where the central bank's mandate is clear and transparency is integral to the framework. Deb et al., 2023 document significantly stronger output and price responses in flexible exchange rate regime countries and in countries with higher levels of financial development. They also show that the credibility and transparency of a central bank affect transmission effectiveness, particularly on prices, due to their role in anchoring inflation expectations.

Figure 1.1. Transmission Channels Examined and Corresponding Financial Indicators



Source: Authors' representation.

Building on these insights and leveraging several years of post-reform data, this paper aims to offer an updated assessment of the effectiveness of the current monetary policy framework and the relative importance of the different transmission channels. We do so by addressing two main questions: First, how strong is monetary policy transmission to output and inflation, and how has it changed since the reforms? Second, which transmission channels are most active today— interest rates, credit, exchange rate, asset prices, and expectations—and how do they differ by instrument or market segment?

To answer these questions, we employ a two-step empirical strategy. First, we identify exogenous monetary policy shocks using a Taylor-rule-like approach (in the spirit of Romer and Romer, 2004), leveraging Bank Al-Maghrib's published forecasts and policy rationales. This method allows us to distinguish genuine policy shocks from systemic responses to current and expected economic conditions—an approach well-suited to Morocco's context of limited high-frequency data.

Second, we estimate the dynamic effects of these shocks using the local projections (Jordà, 2005) method, which is well-suited to settings with limited sample sizes and structural changes.[3] This enables us to trace the impulse responses of output, inflation and key financial variables across both pre-reform (2006Q4-2017Q4) and post-reform (2018Q1-2025Q3) periods. By combining forecast-based identification with flexible estimation

[3] Auerbach and Gorodnichenko (2012) and Ramey and Zubairy (2018) highlight LPs' advantages in capturing nonlinear and state-dependent effects; Jordà and Taylor (2025) explicitly reaffirm LPs' strengths for policy analysis in small samples and evolving regimes. Recent work by Herbst and Johannsen (2020/2024) discusses potential bias in LPs in small samples but also notes that bias correction methods mitigate these concerns.

using local projections, our methodology provides a unified framework for assessing both macroeconomic effects and transmission channels.

Our findings indicate that monetary policy transmission to output and inflation has improved meaningfully in recent years following several reforms. Specifically, a contractionary 1 percentage point monetary policy shock leads to a decline in output and inflation of about 0.3 and 0.12 percent, respectively, with effects peaking after 7 and 8 quarters, compared to declines of about 0.1 and 0.05 percent, respectively, in the pre-reform period. Despite the marked improvement, the responses remain relatively delayed and muted compared to other emerging markets—where output typically declines by 0.3-0.8 percent within 2-6 quarters and inflation by about 0.2-0.3 percent within 4-8 quarters (Willems, 2018; Brandao-Marques et al., 2020; Deb et al., 2023; Checo et al., 2024)—reflecting that not all transmission channels are fully effective.[4]

Regarding the transmission channels, our analysis reveals distinct patterns. The interest rate channel is the most active, in line with previous studies and as in many emerging economies with bank-based financial systems, with policy rate changes passing through to government securities (especially short-term), deposit rates, and bank lending rates by purpose and institutional sector—ranging from 50 to120 percent within 1 to 2 quarters. Pass-through to deposit rates is incomplete—about 60 percent within one to two quarters—likely reflecting structural factors including abundance of low-cost deposits and limited investment alternatives. The credit channel is partially active, with statistically significant declines in short-term and services sector credit following rate hikes, while long-term and agricultural credit show insignificant responses, likely due to fixed-rate structures and subsidized lending or government support schemes. The exchange rate channel has strengthened post-reforms, with the nominal effective exchange rate and the EUR/MAD bilateral rate appreciating following policy tightening. Given capital controls and a still band-based exchange rate arrangement, exchange rate responses may reflect changes in domestic financial conditions and portfolio allocation rather than arbitrage-based uncovered interest rate parity (UIP) dynamics. The asset price channel remains largely inactive, with no significant response of stock market inflation to monetary policy shocks—reflecting the relatively shallow and illiquid nature of Morocco's capital markets and limited household exposure to equity investments. The expectations channel shows modest declines in inflation expectations after policy tightening, partly driven by actual disinflation, suggesting expectations are somewhat reactive and highlighting scope to further strengthen central bank communication and education.

As Morocco advances towards adoption of an inflation targeting regime, in which the policy interest rate would be the main instrument to achieve the inflation target, understanding the dynamics of monetary policy transmission is both timely and relevant for policymakers. This paper aims to provide a clearer picture of the monetary policy transmission in Morocco, to inform the calibration of policy instruments and guide complementary reforms to strengthen transmission.

The remainder of the paper is organized as follows: Section 2 details the recent institutional and operational reforms; Section 3 presents stylized facts and trends; Section 4 assesses the transmission mechanism empirically; Section 5 discusses policy implications and concludes.

[4] The wide range of estimated responses across EMs reflects the heterogeneity in monetary policy frameworks, inflation composition—including the share of food and administered prices, financial depth, and the role of the expectations channels. Morocco's estimated responses fall in the lower-to-middle range of EM estimates.

Table 1.1. Previous Empirical Studies on Monetary Policy Transmission in Morocco

Study (Authors, Year)	Sample Period	Channels Analyzed	Methdology	Main Findings
Queyranne et al. (2021)	2007-2020 *(included the first widening of the ER band of +/- 2.5%)*	Overall transmission to output and inflation	Taylor-rule, LP	* Muted impact on output/inflation; limited ER flexibility * Output: 0.14 pps decline after 10 quarters; inflation: 0.03 pps decline after 8 quarters. * Accounting for the exchange rate channel, output: 0.18 pps decline after 9 quarters; inflation: 0.04 pps decline after 2 quarters.
Bennouna, Chmielewski & Doukali (2019)	2010–2016 *(pre-2018; no ER flexibility)*	Credit (firm-level)	Panel regression	Transmission stronger for small firms; firm characteristics matter
Bennouna (2018)	2006–2017 *(pre-2018; no ER flexibility)*	Lending rates	Panel ECM	Limited, slow pass-through; heterogeneity across banks/products
Bennouna & Bounader (2018)	2007–2017 *(pre-2018; no ER flexibility)*	Interest rate (market rates)	SVAR	Strong transmission to short-term yields; weaker for longer maturities
Bennouna, Lahlou & Mossadak (2016)	2007–2015 *(pre-2018; no ER flexibility)*	Interest rate, credit, asset price	SVAR, NKM	Interest rate & credit channels most effective; muted output/inflation response. Results for output and inflation were not statistically significant.

Source: Authors' summary

2. Recent Institutional and Operational Reforms Supporting Monetary Policy Transmission

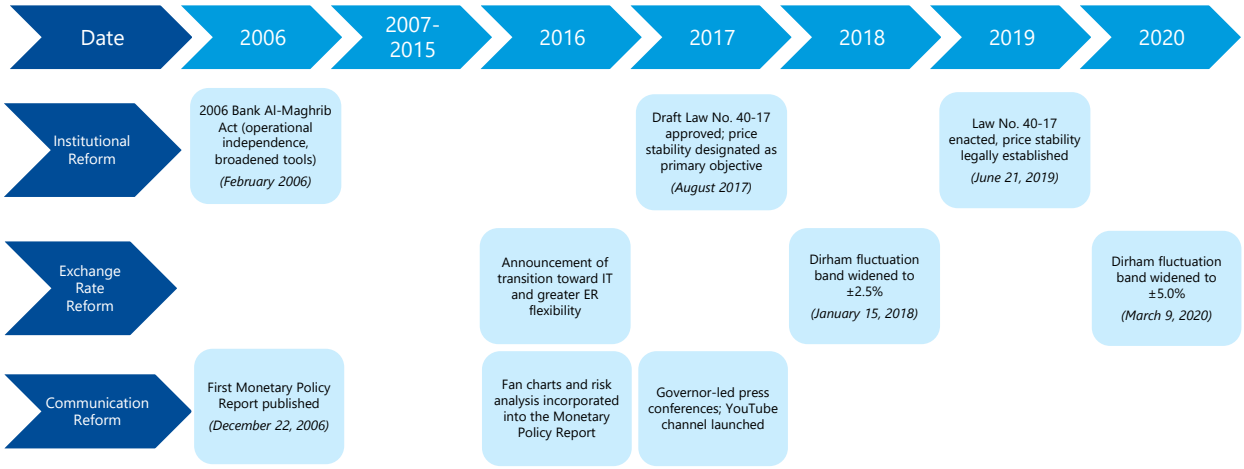
Institutional and operational reforms over the past decade have significantly reshaped Morocco's monetary policy framework, with important implications for the effectiveness of monetary policy transmission. These changes (Figure 2.1)—ranging from establishing price stability as the primary objective, to introducing greater exchange rate flexibility, and enhancing communication—were implemented as part of Morocco's gradual transition toward an inflation-targeting regime and were designed to strengthen the credibility of Bank Al-Maghrib (BAM) and improve the functioning of the key transmission channels. This section reviews these reforms and discusses how they may have influenced the responsiveness of output, inflation, and financial variables to policy actions.

2.1. Price Stability as the Primary Objective (2017–2019)

Designating price stability as the primary objective marked a pivotal step in strengthening Bank Al-Maghrib's policy framework. While the 2006 Bank Al-Maghrib Act ensured operational independence and broadened policy tools, ambiguity remained regarding the hierarchy of objectives. In August 2017, Draft Law No. 40-17 was approved by the Council of Government, reinforcing BAM's independence and formally establishing price stability as its main goal. This reform was enacted through Article 6 of Law No. 40-17, promulgated by Dahir No. 1-19-82 on June 21, 2019 (Bank Al-Maghrib, 2019). The statute states that "the main objective of the Bank is to ensure price stability", while allowing support for growth and other objectives only in the absence of

inflationary pressures.[5] By clearly defining price stability as the overriding objective, this reform likely enhanced the credibility of monetary policy and improved its effectiveness by reducing the uncertainty around the Bank's reaction function. International experience shows that a clear price stability mandate is a prerequisite for successful inflation targeting and effective monetary transmission (Mishkin and Schmidt-Hebbel, 2002; Zelmer et al., 2002; Roger and Stone, 2005). Empirical evidence further demonstrates that well-anchored inflation expectations and high central bank independence enhance the effectiveness of monetary policy, allowing price stability to be restored with lower output costs (Bems et al. 2020; IMF, 2018).

Figure 2.1. Timeline of Key Reforms in Morocco's Monetary Policy Framework



Source: Authors' compilation based on BAM and IMF reports.

2.2. Greater Exchange Rate Flexibility (2018–2020)

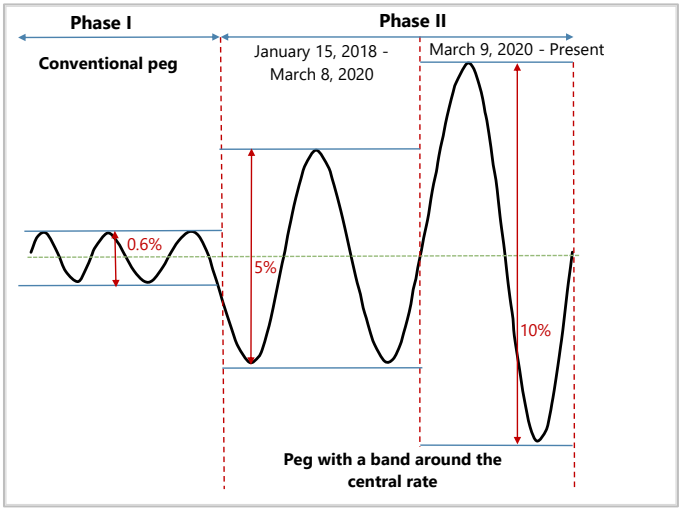
For decades, Morocco maintained a fixed exchange rate regime, using the dirham as a nominal anchor. Initially tied to a broad basket of currencies, the composition was streamlined in 2001 to 80 percent euro and 20 percent US dollar, reflecting trade patterns. In 2015, weights were revised to 60 percent euro and 40 percent dollar. While this framework provided stability, it constrained monetary policy autonomy and limited the effectiveness of monetary policy transmission.

In 2016, BAM announced its intention to transition toward greater exchange rate flexibility and an inflation-targeting regime to strengthen resilience to external shocks and enhance competitiveness. The reform began on January 15, 2018, with the widening of the dirham's fluctuation band from ±0.3 percent to ±2.5 percent around a central rate based on the euro-dollar basket. A second widening of the band, to ±5 percent, followed on March 9, 2020 (Figure 2.2). These steps marked a gradual shift toward a more market-based exchange rate regime, potentially improving the transmission of monetary policy by activating the exchange rate channel and allowing interest rate adjustments to play a greater role. International experience confirms this effect: emerging markets that introduced greater exchange rate flexibility observed a stronger role for the exchange rate channel

[5] https://www.bkam.ma/en/Monetary-policy/Institutional-framework/Objectives

and improved monetary policy transmission (Brandão-Marques et al., 2020; Deb et al., 2023; IMF, 2023; Elkou, 2023; Checo, Grigoli, and Sandri, 2024).

Figure 2.2. Transition Phases of the FX Arrangement



Sources: Bank Al-Maghrib; authors' representation.

2.3. Enhanced Transparency and Improved Communication

BAM's communication strategy has evolved markedly over time, in line with its strengthened price stability mandate and gradual transition toward inflation targeting. While the Bank has published press releases covering its monetary policy decisions alongside its quarterly monetary policy report since late 2006, the last decade has seen major enhancements. These include the introduction of fan charts and risk assessments in the quarterly reports, which are widely used in inflation-targeting frameworks to visualize uncertainty and signal risk awareness[6], to better convey forecast uncertainty, laying the foundation for more forward-looking communication. Around 2017, BAM expanded its communication framework by introducing Governor-led press conferences streamed on social media immediately after Board meetings, complementing the same-day publication of the press release and the monetary policy report. Outreach was also further broadened through educational materials and active engagement on social media, including the launch of BAM's YouTube channel featuring explanatory videos and live streaming of press conferences. These initiatives reflect a clear effort, with evidence of growing engagement over time[7], to improve public understanding and dissemination of monetary policy decisions, and to support the anchoring of inflation expectations and policy credibility.

The IMF's 2022 Central Bank Transparency Code Review for Morocco[8] confirms that these reforms have resulted in a high level of transparency regarding monetary policy and risks to price stability. The review

[6] Fan charts and risk analysis are integral parts of monetary policy communication in inflation targeting countries as the former help to visualize uncertainty, which helps anchoring expectations more realistically and reduce misinterpretation of policy intentions, while the latter signals the central bank's awareness of potentials shocks and contingency planning. Together, these tools strengthen the expectations channel and support forward guidance.

[7] Engagement with BAM's communication has grown significantly, as reflected in a tenfold increase in views of the Governor-led press conferences following the Board's decision between 2017 and 2025 on its official YouTube channel (http://www.youtube.com/@BankAlMaghribBKAM).

[8] IMF Country Report No. 22/380.

classified the transparency of the monetary policy objective and framework as comprehensive, policy decisions as expanded, and supporting analysis as comprehensive. It highlighted BAM's clear explanation of how monetary policy is designed and implemented in view of the price stability mandate stipulated by law, and noted that the January 2018 move to greater exchange rate flexibility and the enactment of the new BAM statute in July 2019 provided momentum for setting a high level of transparency. The review concluded that BAM's transparency framework is well established, covering policy decisions, supporting analysis, monetary operations, and related reporting, with high standards for timeliness, periodicity, and disclosure. Further enhancements, such as publishing Board meeting minutes or summaries of policy discussions, were suggested to further strengthen the already high level of transparency. These achievements place BAM among the more transparent central banks globally and align closely with international best practices.

This alignment with global standards is important, as international evidence supports the importance of these reforms: clear, consistent, and engaging communication is considered a policy instrument in its own right, influencing expectations and reducing uncertainty, and hence strengthening the transmission of monetary policy (ECB, 2021; BIS, 2024). Recent studies show that transparency and credible communication enhance the effectiveness of monetary policy, with more transparent central banks achieving stronger and more timely impacts in prices (Brito et al., 2018; IMF, 2018; Deb et al., 2023). BAM's efforts in this area have likely contributed to better anchoring of expectations, enhanced credibility, and improved signaling of monetary policy. In parallel, BAM strengthened the day-to-day implementation of monetary policy through improvements in liquidity forecasting and the calibration of open market operations; Section 3 documents the associated evolution of short-term market rates, in addition to other stylized facts—including the structure of the financial system and the CPI basket—that shape the transmission to output and inflation.

3. Stylized Facts: Structural Features and Pre- and Post-Reform Operating Environment

This section presents stylized facts that characterize Morocco's evolving monetary policy operating environment and the structural features that may shape the strength and timing of monetary policy transmission. We highlight changes in the operating framework and short-term market rates, the bank-based nature of the financial system, the post-2018 exchange rate environment, and inflation structure and expectations. These descriptive patterns provide context for the empirical assessment in Section 4.

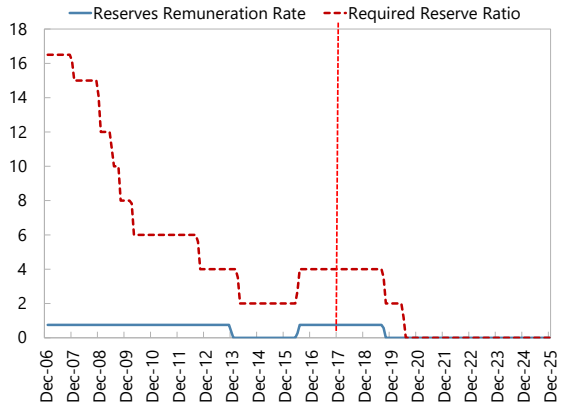
3.1. Monetary Policy Operational Framework and Short-Term Market Rates

A key development in Morocco's monetary policy operational framework has been the shift toward a clearer short-rate anchor, with the policy rate serving as the main signal of the monetary stance and overnight money market rates acting as the first step in transmission. Historically, while the policy rate was in place, BAM relied more heavily on a broad set of liquidity management tools—including reserve requirements, reserves remuneration, and open market operations with varying objectives. Over time, the roles of reserve requirements and reserve remuneration were gradually reduced and ultimately brought down to zero (Figure 3.1), and open market operations became more narrowly focused on steering overnight rates in line with the policy rate. This reinforced reliance on the policy rate as the main operational anchor. Descriptive statistics underscore this shift, with the policy rate adjustments becoming both more frequent and larger in magnitude in

the post-2018 period (Table 3.1): the average time between rate changes declined from 5.6 to 2.5 quarters, while the average absolute rate change increased from 0.13 to 0.40 percentage points per year and the average move size rose from 0.25 to 0.38 percentage points.

Figure 3.1. Reserve Requirements and the Shift Away from Quantity-Based Instruments

(Percent)



Sources: Haver and BAM.

Table 3.1. Policy Rate Setting: Frequency and Magnitude of Policy Rate Adjustments

	Avg duration between rate changes	Avg absolute change per year 1/	Avg absolute change per move	Mean	SD	Min	Max	# of policy rate changes	# of observations
	(in quarters)	*(in percentage points)*		*(in percent)*	*(in percent)*	*(in percent)*	*(in percent)*		
Policy Rate									
Full Sample	3.9	0.24	0.32	2.65	0.58	1.50	3.50	14	75
2006Q4 - 2017Q4	5.6	0.13	0.25	2.94	0.41	2.25	3.50	6	45
2018Q1 - 2025Q3	2.5	0.40	0.38	2.21	0.53	1.50	3.00	8	30

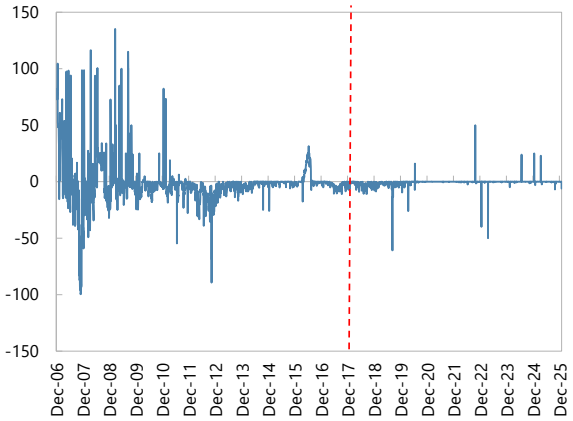
1/ Average absolute change per year is computed as the sum of absolute changes in the policy rate within the subsample divided by the length of the subsample in years.

Sources: BAM and authors' calculations.

This shift was accompanied by a notable improvement in the alignment between the policy rate and overnight interbank rate. Prior to the strengthening of the operational framework, the spread between the policy rate and the overnight rate was highly volatile, with episodes of persistent deviations during periods of surplus liquidity. A recent example was in 2016 when excess liquidity coincided with overnight rates drifting below the policy rate, by around 25 basis points for the overnight unsecured interbank rate and around 75 basis points for the overnight repo rate. These deviations reflected BAM's challenges at the time to fully steer liquidity conditions, allowing surplus liquidity to linger in the banking system and weakening the transmission of the policy signal. As operational enhancements to liquidity management and forecasting progressed, this relationship appears to have improved significantly. With better calibrated open-market operations and stronger liquidity management, the spread between the policy rate and the interbank rate became markedly more stable, and the co-movement between the two rates increased (Figure 3.2). Volatility in the spread diminished sharply after 2018 as BAM maintained tighter control over liquidity conditions, helping to keep overnight rates closer to the policy rate.

Figure 3.2. Policy Rate–Overnight Rate Spread

(Percentage points, policy rate less interbank money market rate)

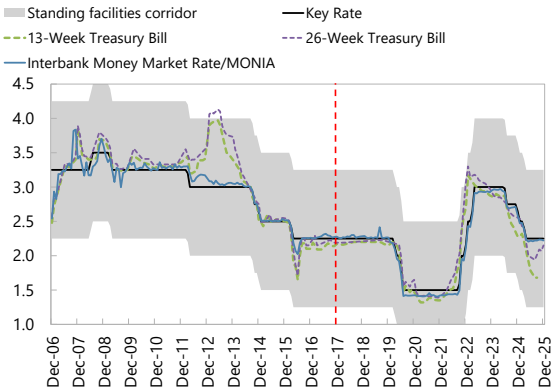


Sources: Haver, BAM, and authors' calculations.

This stronger anchoring of short-term rates also improved transmission along the front end of the yield curve: yields on short-term government securities—particularly 13-week and 26-week Treasury bills—have moved closely with the policy rate and overnight rates in recent years, indicating a clearer and more consistent interest rate structure across short-term money and government securities markets (Figure 3.3). Overall, these patterns suggest that monetary policy implementation has become more predictable at the short end, providing conditions for stronger and more coherent interest rate transmission through money markets and short-term government securities.

Figure 3.3. Short-Term Interest Rates and the Standing Facilities Corridor

(Percent)



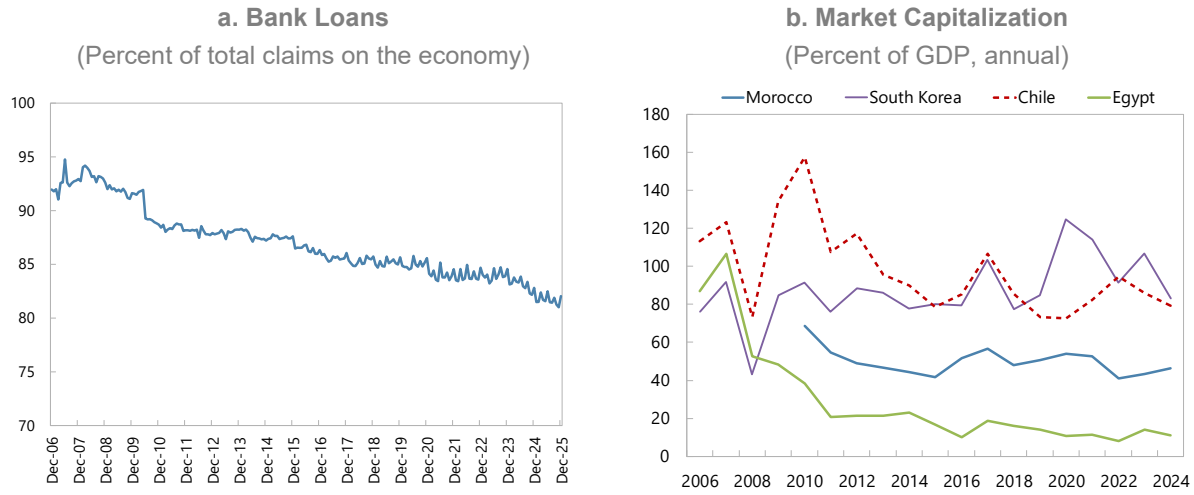
Sources: Haver and BAM.

Note: The overnight interbank rate is shown up to 2019; from 2020 onward, the MONIA (Moroccan Overnight Index Average) is used following its introduction by BAM.

3.2. Financial System Structure and Developments

Morocco's financial system is characterized by the dominance of banks and limited capital market depth, with bank loans accounting for over 80 percent of total claims on the economy, despite a gradual decline over time (Figure 3.4a).[9] In such a system, monetary policy transmission is expected to operate mainly through bank funding and lending conditions—through deposit and lending rates and the availability of bank credit—while market-based channels linked to broad asset price developments are likely to play a more limited role.

Figure 3.4. Bank Loans and Market Capitalization in Morocco



Sources: BAM, Haver, World Federation of Exchanges, and World Bank Group.

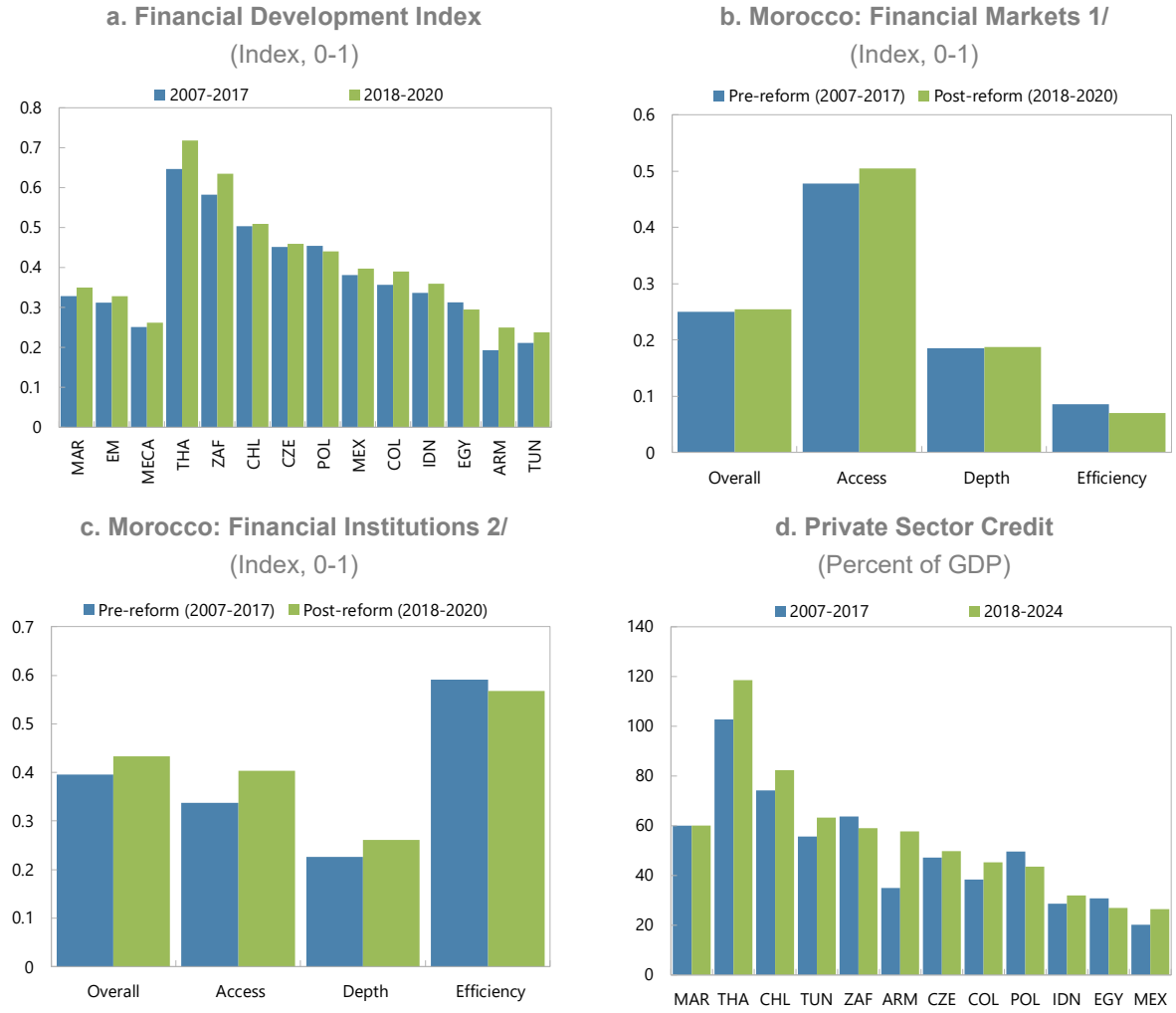
Cross-country evidence suggests that stronger financial development can amplify monetary transmission by broadening the set of instruments and markets through which policy affects financing conditions (Deb et al., 2023; IMF, 2023; Checo et al., 2024). In this context, Morocco has made some, though relatively modest, progress in financial development and deepening over the past decade. Indicators based on the IMF Financial Development Index place Morocco close to the emerging-market average and ahead of regional peers, although still well below more financially advanced emerging markets. The observed gains are limited and uneven—mainly reflecting improvements in access—while market depth and efficiency remain relatively low (Figure 3.5, Panels a-c).

Developments in private sector credit provide complementary evidence on financial depth and the potential role of the credit channel. Morocco's private credit-to-GDP has remained broadly stable over time but continues to be relatively high by regional standards, though it remains below levels observed in more financially advanced emerging markets such as Chile and Thailand (Figure 3.5d). Market capitalization as a share of GDP remains modest relative to these peers, reinforcing the limited scope for asset price and wealth channels (Figure 3.4b). Overall, these features point to a transmission mechanism that relies primarily on the banking system and

[9] Total claims on the economy refer to the stock of financial claims held by the domestic financial system on households, non-financial firms, and the government, including bank credit and holdings of domestic debt instruments.

short-term interest rates, with financial market-based channels likely constrained by the still limited capital market depth.

Figure 3.5. Financial Deepening and Market Structure in Morocco



Sources: IMF Financial Development Index, IMF International Financial Statistics, and IMF World Economic Outlook.
1/ Financial markets=capital markets (stocks, bonds, securities); depth=size and scale of financial system; access=ability of users to obtain financial services; efficiency=how well resources are allocated and services delivered.

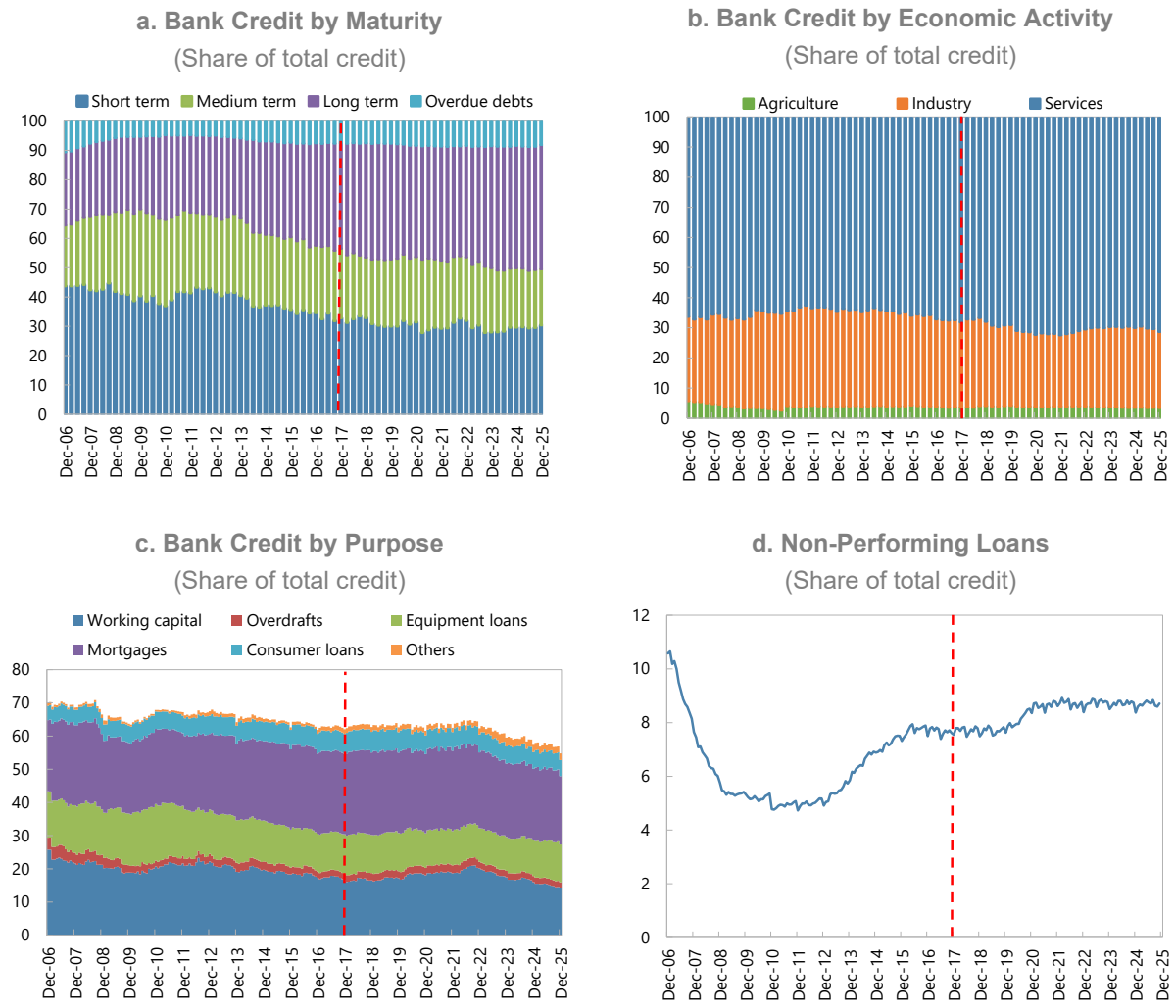
2/ Financial institutions=financial intermediaries (banks, insurers, etc.); depth=size and scale of financial system; access=ability of users to obtain financial services; efficiency=how well resources are allocated and services delivered.

Note: Data on post-reform period available from 2018-2020.

3.3. Banking Sector: Credit Features Shaping Transmission

Within a bank-based system, the composition of bank credit would shape how quickly policy rate changes translate into lending conditions and credit volumes. In particular, the maturity structure of the credit portfolio can affect the repricing frequency of loans: short-term facilities, including working capital and overdrafts, typically reprice more frequently, and therefore tend to show stronger and faster pass-through, while longer-term or capped products, (including some payroll-deducted loans) adjust more slowly and incompletely (Leigh, 2025). In Morocco, the maturity composition of credit has shifted over time toward longer maturities, while the share of short-term credit has declined (Figure 3.6a). While this shift is broadly consistent with financial deepening and greater availability of longer-term financing, longer-maturity loans tend to reprice less frequently than short-term facilities. As a result, the increasing share of longer-term credit in the outstanding stock of credit may contribute to a more gradual adjustment in aggregate credit conditions following changes in the monetary policy stance.

Figure 3.6. Bank Credit Composition and Asset Quality in Morocco



Sources: BAM and authors' calculations.

The allocation of credit across economic activities can further shape transmission, as sectors differ in interest sensitivity, typical loan maturities, and exposure to demand versus supply shocks. In Morocco, bank credit is distributed across services, industry (including manufacturing and construction), and agriculture (Figure 3.6b). Credit to agriculture may be influenced by different government support schemes and seasonal factors, while industry credit often reflects medium and long-term investment financing, both of which can imply less frequent repricing. In contrast, services-related credit is often more closely linked to working-capital needs and shorter maturities. This sectoral allocation suggests that aggregate credit conditions may adjust unevenly across activities, contributing to heterogeneous and potentially delayed aggregate responses.

Credit composition by purpose provides additional insights into interest-rate sensitivity. In Morocco, working capital and overdraft facilities—which tend to be shorter maturity and more frequently repriced—have declined as a share of total credit over the past decade and represent about one quarter of total credit, whereas longer-term lending, including investment loans and mortgages, remains sizable (Figure 3.6c). This composition suggests that changes in the policy stance may pass through more quickly to some lending categories than others, contributing to heterogenous and potentially delayed aggregate credit responses. Finally, asset quality can influence banks' willingness to extend credit by increasing perceived risk and tightening lending standards. Nonperforming loans have risen over the last decade (Figure 3.6d), partly driven by the Covid-19 pandemic shock, which may affect lending conditions and risk premia and in turn dampen the response of credit to changes in the monetary policy stance. Overall, these credit portfolio features—maturity, sectoral allocation, purpose and asset quality—provide useful structural context for interpreting the empirical analysis of the credit channel in Section 4.

3.4. Exchange Rate Regime and Flexibility

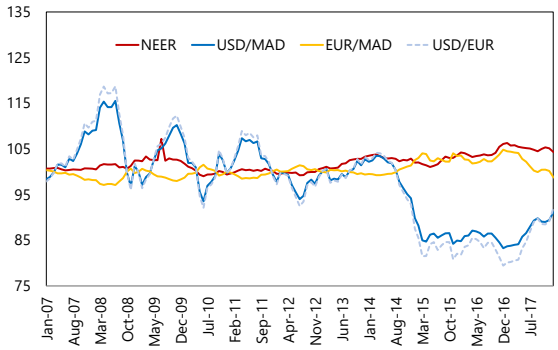
Morocco's move toward greater exchange rate flexibility increased the scope for exchange rate adjustment and reshaped the environment for monetary policy transmission. Following the widening of the dirham's fluctuation band in 2018 and again in 2020, exchange rate movements have become more variable, creating conditions under which the exchange rate channel can play a more meaningful role. The behavior of the nominal effective exchange rate (NEER) and key bilateral rates illustrates the change in the exchange rate environment. Pre-reform, NEER and EUR/MAD were remarkably stable, consistent with a tightly managed basket peg and narrow fluctuation band, while USD/MAD was volatile and largely mirrored global dollar-euro movements (Figure 3.7a). Post-reform, NEER shows a clear upward drift and wider two-way movement, while EUR/MAD exhibits more pronounced moves, consistent with improved price formation and greater scope for exchange rate adjustment under wider bands. In contrast, USD/MAD continues to move closely with USD/EUR, reflecting global dollar cycles rather than domestic conditions (Figure 3.7b). This increase in variability is also visible in the distribution of exchange rate changes: box-and-whisker plots indicate a wider dispersion and more pronounced two-way movements in daily exchange rate changes post-reform, particularly for the EUR/MAD rate (Figure 3.7c).

This increase in exchange rate volatility and the more dynamic behavior of the EUR/MAD rate suggest that the exchange rate channel is now more plausibly operative in Morocco's monetary transmission mechanism. Greater flexibility allows the exchange rate to respond to monetary policy shocks and external developments, potentially strengthening the transmission to output and prices. However, the effectiveness of the exchange rate channel depends not only on exchange rate flexibility but also on FX market depth, expectations, and the

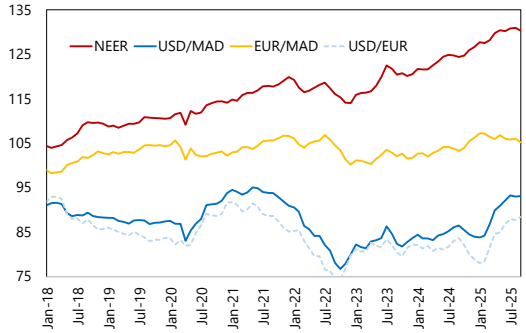
degree of pass-through to domestic prices. While capital controls have been relaxed gradually and measures of financial openness point to relatively high formal openness—particularly for non-resident outflows—limited FX and capital market depth, together with a relatively narrow set of liquid debt and equity investment opportunities, constrain the scope for interest-rate driven cross-border portfolio reallocation (Figure 3.8). This, however, does not necessarily preclude exchange rate adjustment through domestic financial and balance sheet channels such as portfolio shifts, liquidity conditions, and expectations. Overall, these developments suggest that Morocco's monetary policy framework now provides conditions for a more active exchange rate channel, though its quantitative importance is assessed in Section 4.

Figure 3.7. Exchange Rate Behavior Before and After Reform

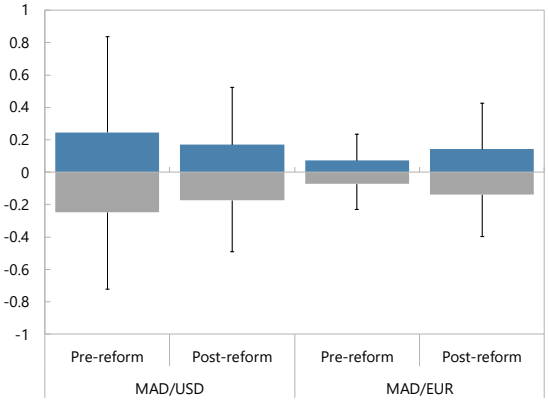
a. NEER and Bilateral Rates: Pre-Reform
(Index; 2010=100)



b. NEER and Bilateral Rates: Post-Reform
(Index; 2010=100)



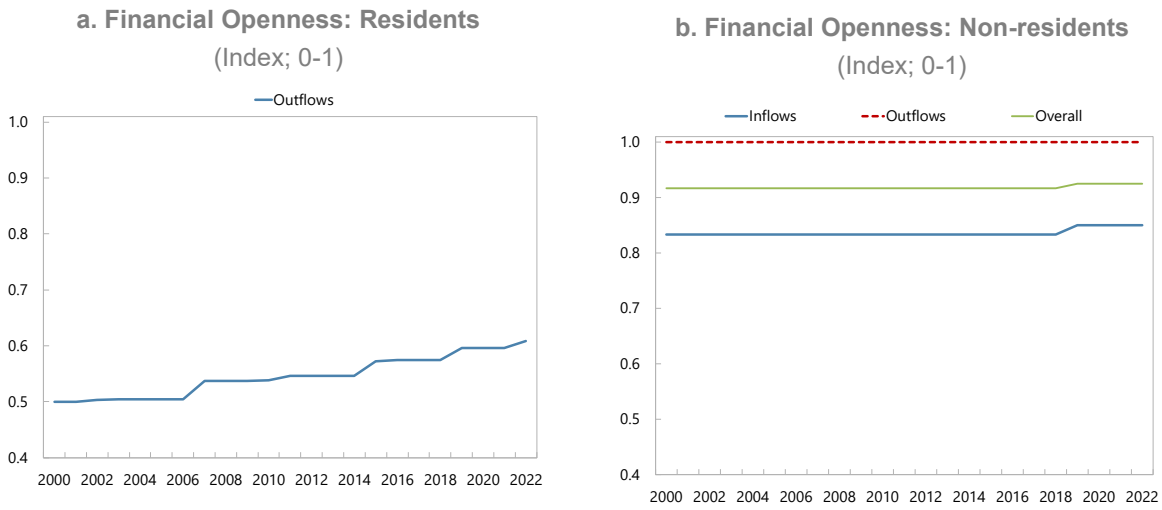
c. Exchange Rate Volatility: Pre- and Post-Reform
(Percent)



Sources: Bank Al-Maghrib, Haver, and authors' calculations.

Note: Panel (c) reports box-and-whisker plots of daily percentage changes in bilateral exchange rates before and after the reform. The boxes denote the interquartile range (25th–75th percentiles), the center line indicates the median, and the whiskers correspond to the 5th and 95th percentiles.

Figure 3.8. Financial Openness



Source: Li (2026).

Note: Higher values indicate greater capital openness. For residents, the openness index reflects controls on capital outflows only, as resident inflows are not applicable. For non-residents, indices are shown separately for inflows, outflows, and overall openness.

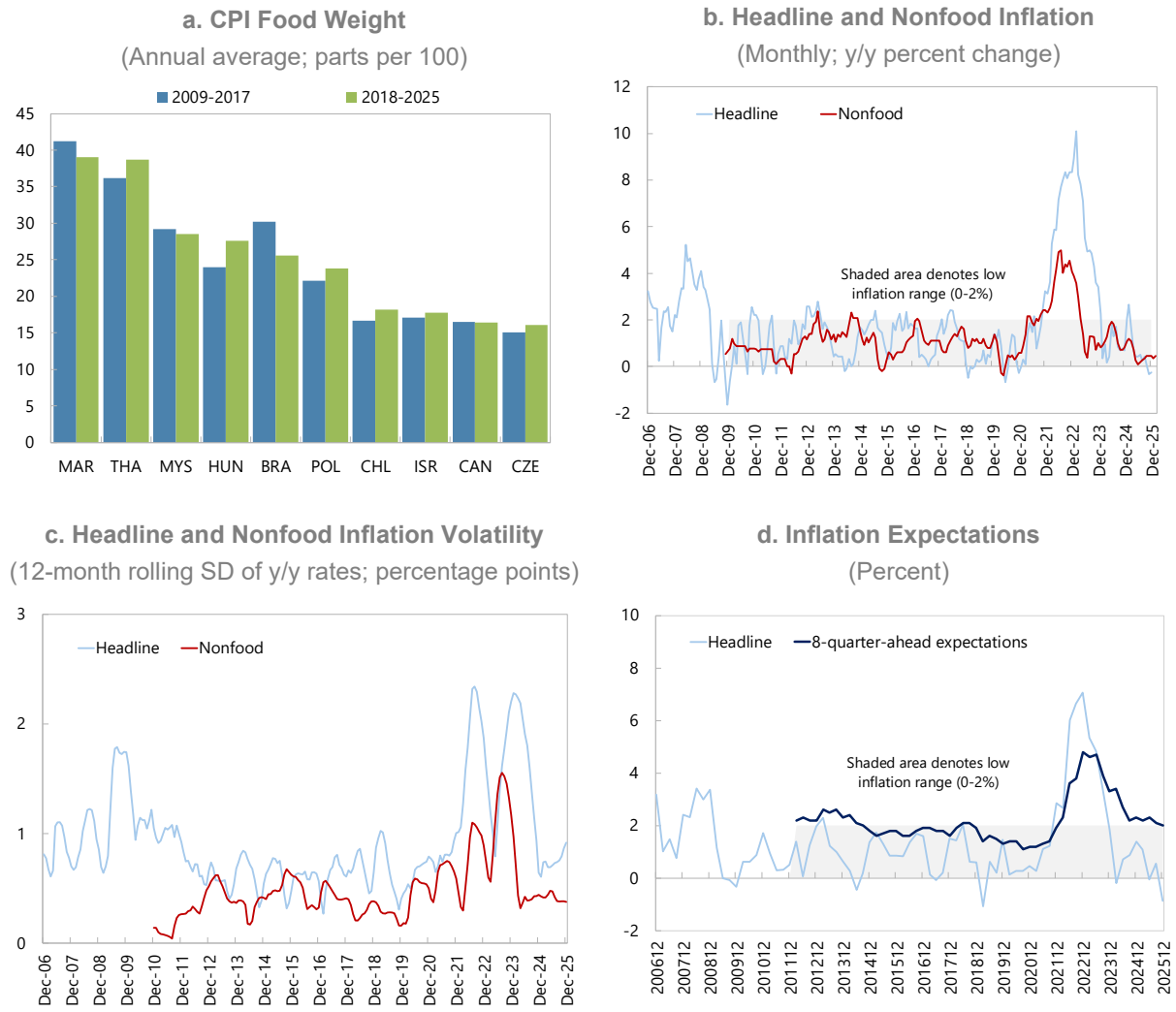
3.5. CPI Basket and Price Rigidity

Morocco's inflation dynamics shape the transmission of monetary policy to prices. The CPI basket has a large share of food and regulated prices—about 40 percent and 20 percent, respectively—substantially larger than in many other emerging market and advanced economies (Figure 3.9a). These components tend to be less responsive to domestic demand conditions and interest rate changes, as their pricing is influenced more by administrative decisions and supply-side factors, including weather shocks and import prices. As a result, changes in the policy stance are more likely to affect inflation gradually.

Inflation has also been relatively low and stable over time, with headline and nonfood inflation moving within a narrow range for extended periods outside major shock episodes (Figure 3.9b). Consistent with this pattern, inflation volatility—measured as the rolling 12-month standard deviation of y/y inflation—has generally remained contained, particularly for nonfood inflation, and rises mainly during large shock episodes (Figure 3.9c). This low-inflation, low-volatility environment likely reflects the effectiveness of Morocco's monetary and exchange rate framework as a nominal anchor, which helped contain second-round effects and limit the propagation of shocks. While these conditions support price stability, they also imply that inflation responds only gradually to monetary policy actions, as wage-price feedback and demand-driven amplification mechanisms tend to be muted. Inflation expectations provide further insights into the inflation environment. The 8-quarter-ahead inflation expectations remained broadly aligned with low inflation over most of the sample and adjusted only gradually—even during the 2021-2022 inflation surge—consistent with well-anchored expectations under a stable nominal anchor (Figure 3.9d). At the same time, the gradual adjustment of expectations indicates that they are largely adaptive to realized inflation outcomes rather than strongly forward-looking with respect to short-term policy actions.

Overall, the combination of CPI basket composition, sustained inflation stability, and anchored expectations is consistent with gradual and relatively muted transmission of monetary policy to inflation. The role of these structural features is empirically examined in Section 4.

Figure 3.9. Inflation Structure and Dynamics



Sources: Haver, IMF International Financial Statistics, HCP and authors' calculations.
Note: Nonfood inflation excludes food items but includes administered/regulated nonfood prices; therefore, it should not be interpreted as a pure measure of underlying inflation pressures. Core measures available for Morocco still include a sizable food component and hence can co-move with headline inflation during food-price shocks.

Overall, these stylized facts point to a transmission environment characterized by effective control of short-term market rates, a bank-dominated financial system with heterogeneous repricing across loan maturities and segments, and inflation dynamics shaped by the composition of the CPI basket and a policy framework that has historically provided a stable nominal anchor, particularly during low-inflation periods. These features suggest that monetary policy effects may be uneven across channels and more muted for inflation. Section 4 formally quantifies these patterns using a unified empirical framework.

4. Assessment of Monetary Policy Transmission: Methodology and Empirical Results

4.1. Data and Sample Design

This section quantifies the transmission of monetary policy to macroeconomic outcomes and key channels using quarterly data for Morocco over 2006Q4-2025Q3.[10] The analysis focuses on (i) the effects of identified monetary policy shocks on output and inflation and (ii) the responses of financial indicators that shed light on transmission channels. Given Morocco's evolving policy framework, we examine results in the full sample and in pre- and post-2018Q1 subsamples. The sample split is motivated by the timing of major institutional and operational changes—most notably the move toward greater exchange rate flexibility beginning in January 2018, the strengthened legal focus on price stability between 2017 and 2019, and enhancements in communication and transparency—while emphasizing that the analysis does not seek to quantify the causal impact of reforms on transmission. Instead, the sample split provides a natural context to assess whether estimated transmission dynamics differ across periods.

Our baseline macroeconomic outcomes are real activity and prices. Real activity is measured using real GDP, and inflation are measured using CPI. Robustness checks use alternative real activity and inflation measures. Financial variables used to characterize channels include government securities yields at different maturities, bank deposit and lending rates (by purpose and institutional sector), credit aggregates (total, by maturity, and by sector), equity prices, and exchange rate measures (NEER and key bilateral rates). Annex I provides a full list of variables and sources (Table A.1).

4.2. Identifying Monetary Policy Shocks

4.2.1. Identification Challenge

Assessing how monetary policy affects the economy is challenging because policy decisions are often endogenous: policy rates may rise in response to projected inflation or output growth, creating the illusion that higher rates cause higher inflation or stronger activity.[11] To address this risk of reverse causality, the empirical literature typically proceeds in two steps: first, identify monetary policy changes that are exogenous to current and expected economic conditions; second, estimate their dynamic effects on output and prices.

[10] The full sample comprises 75 quarterly observations (2006Q4–2025Q3), while the post-reform subsample includes about 30 quarterly observations. Although relatively short, sample sizes of this magnitude are common in empirical applications of local projections, particularly in emerging market studies where data availability is limited data and structural change is more likely. Herbst and Johannsen (2024) document that, among the 100 most-cited published studies using local projections, the median sample size is around 95 observations, with a substantial share of papers relying on samples of approximately 30 observations. As discussed in their analysis, small samples can widen confidence intervals and introduce attenuation bias, but local projections remain informative for characterizing the qualitative shape, timing, and relative ordering of responses. The implications of small-sample attenuation for the interpretation of LP impulse responses are discussed further in Annex III.

[11] This is a well-documented phenomenon in the VAR literature, commonly referred to as the 'price puzzle', where a contractionary monetary policy shock is followed by rising prices contrary to a decline as theory would predict.

4.2.2. Choice of Identification Strategy in the Moroccan Context

Given Morocco's data environment and evolving policy framework, the identification of exogenous monetary policy shocks poses several challenges. We therefore adopt a narrative, reaction-function approach in the spirit of Romer and Romer (2004), leveraging Bank Al-Maghrib's regularly published forecasts and policy assessments to isolate the unexpected component of policy decisions. This approach is particularly well suited to settings where financial markets are not deep enough to support market-based identification and where structural change may limit the reliability of tightly specified VAR models.[12]

Narrative and reaction-function approaches to identifying monetary policy shocks in the spirit of Romer and Romer (2004) have been widely applied in the monetary transmission literature, particularly in settings where policymakers rely on forecasts and market-based identification is constrained. Country-specific applications include Cloyne and Hürtgen (2016) for the UK, Holm et al. (2020) for Norway, Eklou (2023) for Malaysia, and Lee and Park (2025) for Korea. Cross-country applications include Brandão-Marques et al. (2020) for a broad set of emerging market economies and Tiedemann et al. (2025) for Sub Saharan Africa.[13] Complementary approaches identify monetary policy shocks using forecast-based policy surprises derived from professional surveys, especially in emerging markets where high-frequency financial data are unavailable (Deb et al., 2023; Checo et al., 2024; and Leigh and Xu, 2025).[14]

4.2.3. Baseline Reaction Function and Shock Construction

We identify monetary policy shocks using a Taylor-rule-like reaction function estimated on quarterly data over 2006Q4-2025Q3:

$$\Delta i_t = \alpha_0 + \alpha_1 E_t \Delta y_{t+4} + \alpha_2 E_t \pi_{t+4} + \sum_{j=1}^{2} \alpha_{3j} \Delta y_{t-j} + \sum_{j=1}^{2} \alpha_{4j} \Delta p_{t-j} + \sum_{j=1}^{2} \alpha_{5j} i_{t-j} + \varepsilon_t \qquad (1)$$

where i denotes the monetary policy rate, $E_t \Delta y_{t+4}$ and $E_t \pi_{t+4}$ denotes BAM's 4-quarter-ahead forecasts of real GDP growth and inflation (baseline uses core CPI inflation forecasts); y denotes real output and p denotes prices. Our baseline specification in (1) also includes two quarter lags of i to capture the persistence in the policy rate, and the residuals, ε_t , capture the monetary policy shock. Under this approach, shocks are identified even when the policy rate is unchanged, because deviations from the rule-implied stance capture unexpected shifts in policy relative to macroeconomic conditions. Figure 4.1a shows that the estimated shocks exhibit meaningful variations even in periods with infrequent policy rate changes. We report the coefficients estimates for the baseline and alternative reaction function specifications in Annex II (Table A.2).

To assess robustness, we consider alternative specifications of equation (1) that incorporate additional indicators that may systematically affect monetary policy decisions in a small open economy, including the

[12] Alternative identification strategies—including structural VARs, high-frequency identification based on financial market surprises, and proxy-SVAR or external-instrument approaches—are widely used in advanced economy settings but typically require deep financial markets, high-frequency asset price data, or suitable external instruments that are not readily available for Morocco. See Sims (1980), Bernanke and Blinder (1992), Christiano et al. (1999), Kuttner (2001), Gertler and Karadi (2015), and Stock and Watson (2012, 2018) for representative approaches, and Checo et al. (2024) for survey-based high-frequency adaptation in emerging market contexts.

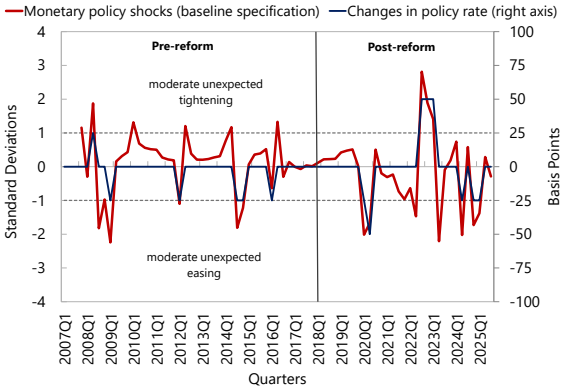
[13] Given the limited availability of central bank forecasts and detailed narrative records across countries, cross-country empirical applications of narrative and reaction-function approaches typically rely on external forecasts (such as Consensus Forecasts) as proxies for policymakers' expectations, rather than central bank forecasts in the strict sense of Romer and Romer (2004).

[14] Deb at el. (2023) uses Consensus Forecasts to derive monetary policy shocks, while Checo et al. (2024) uses Bloomberg survey data, while Leigh and Xu (2025) use the Focus survey maintained by the Central Bank of Brazil.

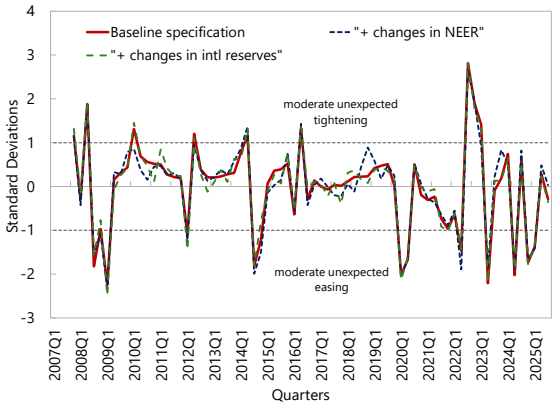
NEER, bilateral exchange rates (USD and EUR), the ECB policy rate, and international reserves.[15] Results are robust to these different specifications (Figure 4.1b).

Figure 4.1. Identified Monetary Policy Shocks

a. Shocks versus Actual Changes
(Standard deviations, unless indicated otherwise)



b. Baseline versus Alternative Specifications
(Standard deviations)



Sources: Haver, BAM, and authors' estimates.

Note: Shocks are residuals from a reaction function using 4-quarter-ahead BAM forecasts.

4.3. Estimation Framework: Local Projections

We estimate the dynamic effects of monetary policy shocks using the local projections (LP) approach of Jordà (2005). This framework is well suited to the Moroccan context, as it is flexible, robust to misspecification, and well suited to settings characterized by structural change. For each outcome variable y_t, we estimate:

$$y_{t+h} - y_{t-1} = \alpha^h + \sum_{k=1}^{l} \gamma_k^h y_{t-k} + \sum_{k=0}^{l} \beta_k^h MP_{t-k} + \sum_{k=0}^{l} \theta_k^h X_{t-k} + \varepsilon_{t+h}^h \quad (2)$$

where y_{t+h} is the outcome variable *h*-periods ahead, and MP_t is the identified monetary policy shock from equation (1). The coefficient β^h captures the impact of monetary policy shocks on the outcome variable over the *h*-period horizon. All specifications include two lags of the dependent variable and the monetary policy shock to capture persistence and dynamic adjustment. The vector X_t includes control variables that may influence macroeconomic and financial conditions in Morocco, such as global commodity prices, and measures of global financial conditions. To account for the exceptional disruption during the COVID-19 pandemic, we include a pandemic control in all regressions. Equation (2) is estimated separately for each horizon h = {0,1, 2, …, 8}, using ordinary least squares. Confidence intervals are constructed using Newey–West standard errors to account for potential serial correlation and heteroskedasticity in the error terms. We estimate the LPs for the full sample (2006Q4-2025Q3) as well as separately for pre-reform period (2007Q1-2017Q4) and post-reform period (2018Q1-2025Q3), where data allow. We focus on pre- and post-2018 comparisons for macroeconomic outcomes, where regime changes are most relevant, while financial indicators are shown using the full sample

[15] The inclusion of other variables like the exchange rate in the Taylor rule-type specification is common in the literature for emerging markets to account for other factors that might influence some central banks when deciding on interest rates (Deb et al., 2023; Brandao-Marques et al., 2020).

unless otherwise noted. Differences across subsamples are interpreted as changes in transmission patterns across regimes rather than as causal effects of specific reforms.

4.4. Macroeconomic Effects of Monetary Policy Shocks

We first assess the dynamic responses of output and inflation to a contractionary monetary policy shock. Figure 4.2 reports impulse responses over horizons up to eight quarters: Panel (a) shows full-sample responses, while Panels (b)–(c) compare pre- and post-2018 subsamples for output and inflation, respectively.

- *Full sample*. A 1 percentage point contractionary monetary policy shock reduces output by about 0.1 percent after 4-6 quarters and lowers inflation by about 0.06 percent after 8 quarters, with inflation responding more slowly than output, consistent with standard transmission patterns.

- *Pre- vs post-2018*. The estimated responses are stronger in the post-2018 period than in the pre-2018 period. In the post-2018 subsample, a 1 percentage point contractionary shock is associated with a larger decline in output (about 0.3 percent, peaking around 7 quarters) and a larger—though still delayed—decline in inflation (about 0.12 percent, peaking around 8 quarters) compared to roughly 0.1 and 0.05 percent in the pre-2018 subsample (Figure 4.2, b-c). Despite the improvement, output and price responses remain relatively muted and materialize with longer lags compared to benchmark estimates for other emerging markets— where output typically declines by 0.3-0.8 percent within 2-6 quarters and inflation by about 0.2-0.3 percent within 4-8 quarters (Willems, 2018; Brandao-Marques et al., 2020; Deb et al., 2023). These patterns are consistent with the stylized facts presented in Section 3, including a bank-based financial system and inflation dynamics shaped by food and administered prices.

Figure 4.2. Output and Price Responses: Full Sample, Pre- and Post-2018

a. Full Sample

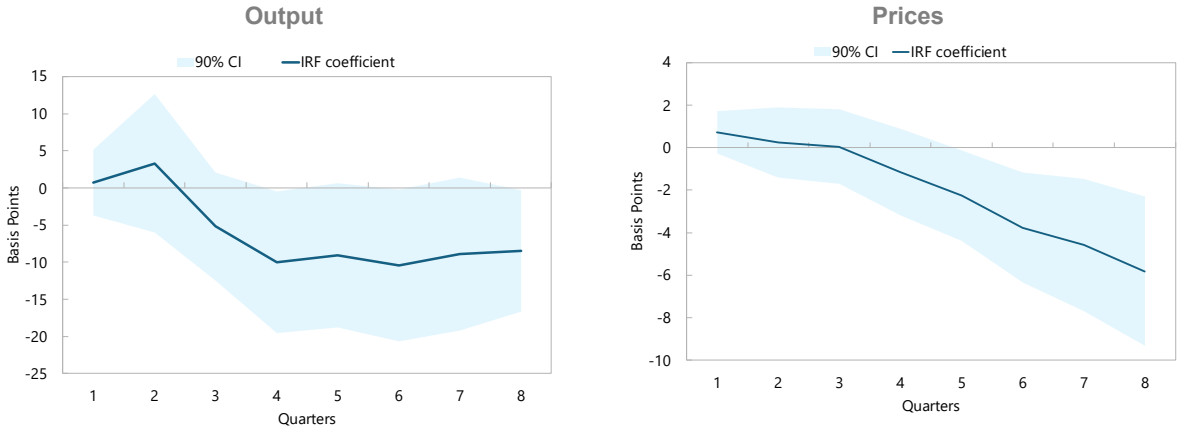


Figure 4.2. Output and Price Responses: Full Sample, Pre- and Post-2018 (concluded)



Notes: Impulse responses are shown for a contractionary one percentage point monetary policy shock. Shaded areas denote 90 percent confidence bands. The pre-reform subsample covers 2006Q4–2017Q4, while the post-reform subsample covers 2018Q1–2025Q3.

4.5. Monetary Policy Transmission Channels

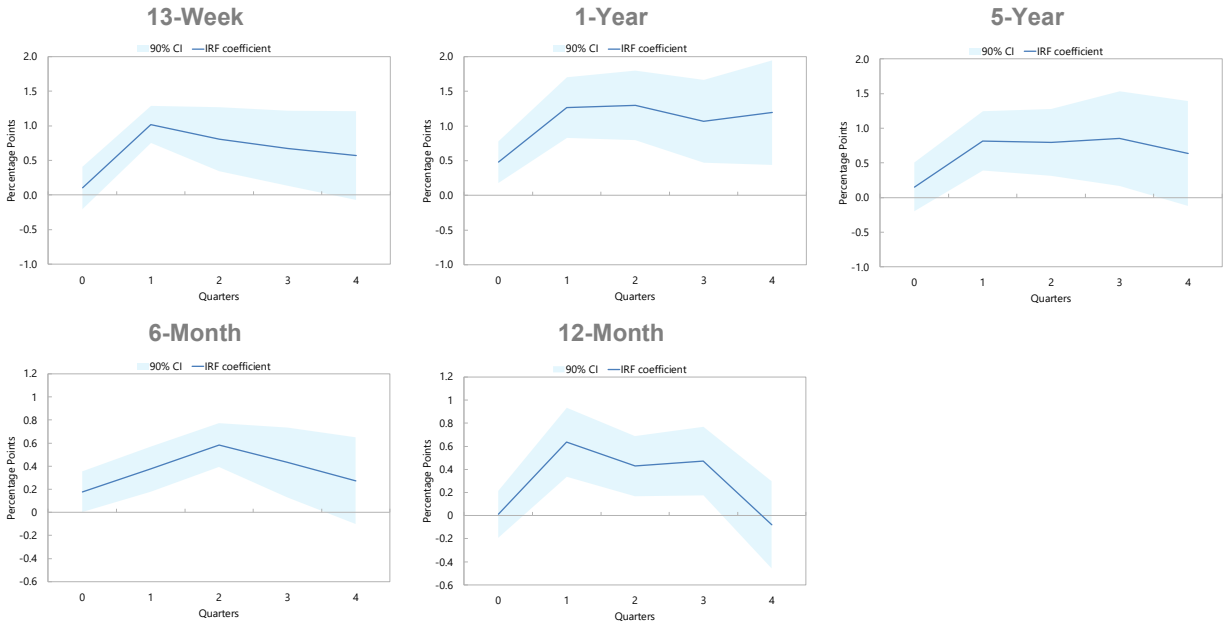
We next examine the responses of key financial indicators to monetary policy shocks to infer the relative strength of transmission channels. Using the local projections specification in equation (2), we report impulse responses for government securities yields, bank deposit and lending rates, credit aggregates, the exchange rate, equity prices, and inflation expectations, over 4-period horizon. Unless otherwise noted, the results in this section are shown for the full sample; where subsample estimates are informative—most notably for exchange rates following the introduction of greater flexibility—we highlight pre- and post-2018 patterns separately.

Overall, the channel evidence below points to strong interest rate pass through, a partially effective and heterogeneous credit channel, an exchange rate channel that operates primarily through domestic conditions under capital controls and band-based exchange rate framework, and weaker asset price and expectations channels.

Interest Rate Channel

Monetary policy shocks are transmitted strongly to short-term money market rates and government securities yields at shorter maturities, consistent with effective short-term rate control and a clear pass-through along the front end of the yield curve. Yields rise in a statistically significant manner across short- and medium-term maturities, with effects concentrated in the first few quarters and stabilizing thereafter (Figure 4.3, top panel). In contrast, long-term yields (10-year bonds, not shown) display weaker and often statistically insignificant responses, consistent with greater sensitivity of long-term rates to risk premia and longer-horizon macro expectations rather than short-term policy shocks.

Figure 4.3. Interest Rate Channel: Government Securities (top panel) and Bank Deposit Rates (bottom panel)

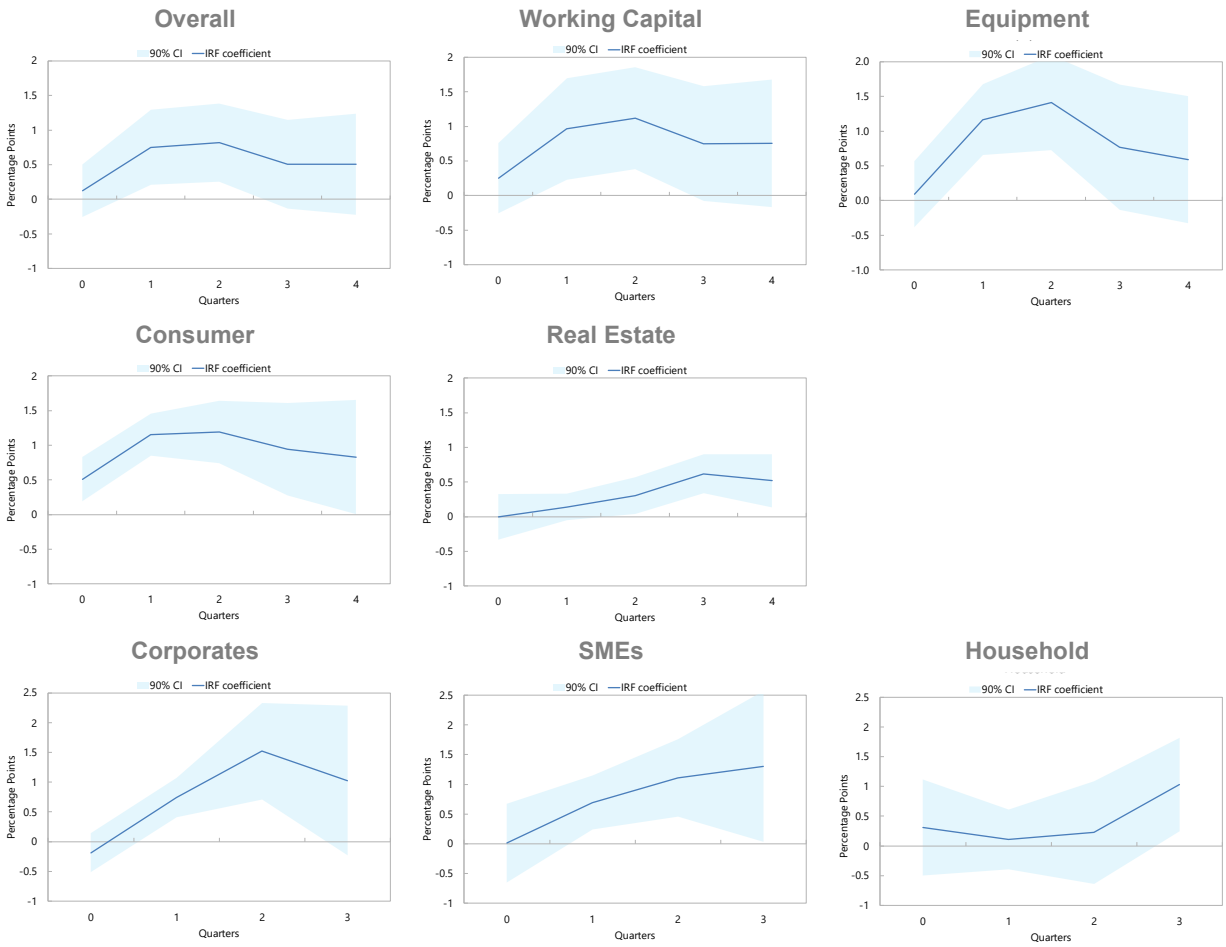


Notes: Impulse responses are shown for a contractionary one percentage point monetary policy shock. Shaded areas denote 90 percent confidence bands. Full-sample results are shown; post-2018 responses are modestly stronger but less precisely estimated.

Pass-through to bank deposit rates are statistically significant but incomplete (Figure 4.3, bottom panel). Deposit rates increase across maturities (6-months and 12-months), with about 60 percent of the shock transmitted within 1-2 quarters in the full sample, consistent with structural features of the banking system (including abundant low-cost deposits and limited alternative savings instruments). Post-2018 estimates are directionally similar and modestly stronger.

Bank lending rates respond positively to monetary policy shocks across all loan purposes, with most categories showing nearly full pass-through within a few quarters—except real estate loans, which exhibit only about 50 percent pass-through after 3 quarters, suggesting slower adjustment in longer-term or collateralized lending (Figure 4.4). Responses by institutional sector suggest faster pass-through for corporates and a more gradual adjustment for SMEs—which may reflect differences in contract structure and pricing practices in addition to the role of public guarantee or support schemes in some segments. Household lending shows the slowest pass-through after 3 quarters.

Figure 4.4. Interest Rate Channel: Bank Lending Rates by Purpose and Institutional Sector

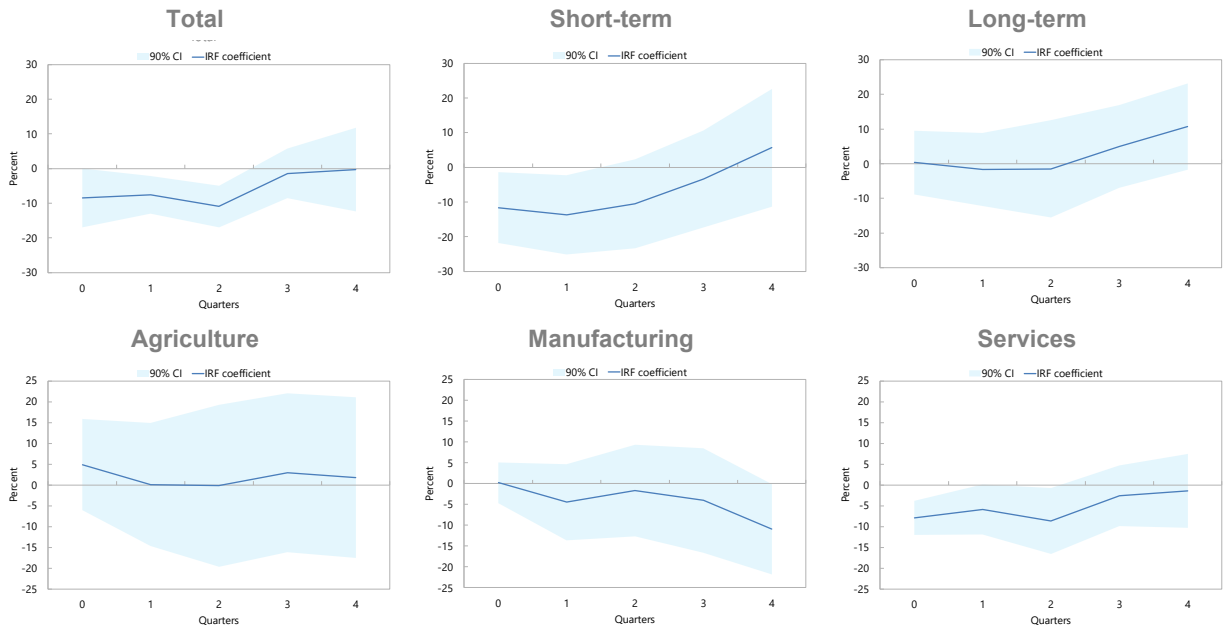


Notes. Impulse responses are shown for a contractionary one percentage point monetary policy shock. Shaded areas denote 90 percent confidence bands. Lending rates by economic purpose are available from 2010Q1; results are therefore estimated over the available sample. Lending rates by institutional sector are available from 2018Q1, so those estimates reflect the post-2018 period only.

Credit Channel

Credit aggregates contract following a contractionary monetary shock, but the response varies by credit maturity and economic activity (Figure 4.5). Short-term credit shows a stronger and more immediate response, highlighting the greater sensitivity of short-term lending to policy changes. In contrast, long-term credit shows a weaker and statistically insignificant response, possibly reflecting fixed-rate contracts, longer repricing horizons, and institutional inertia in the adjustment of longer-term credit. For context, in 2025, short-term loans accounted for about 30 percent of total bank credit, while long-term loans represented around 40 percent (Section 3). This implies that even relatively strong responses in the short-term credit translate into more modest movements in total credit stocks. Sectoral responses further underscore heterogeneity in the credit channel. Credit to services—which tends to rely more on working-capital facilities—shows a clearer contraction following policy tightening, while credit to agriculture and manufacturing exhibits weaker and less precisely estimated responses. These differences are consistent with structural features documented in Section 3, including the role of government support schemes, subsidized lending, and longer average maturities in some sectors.

Figure 4.5. Credit Channel: Credit by Maturity and Economic Activity



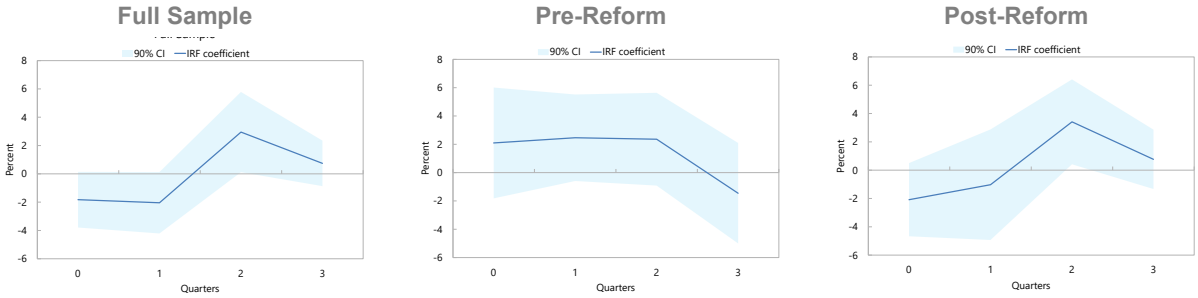
Notes: Impulse responses are shown for a contractionary one percentage point monetary policy shock. Shaded areas denote 90 percent confidence bands. Full sample results are shown.

Exchange Rate Channel

In a small open economy, monetary policy can affect the exchange rate through several mechanisms. In Morocco, however, the exchange rate channel may be viewed as potentially limited by capital controls and by a band-based exchange rate arrangement, which can weaken arbitrage-based uncovered interest parity (UIP) conditions. To address these concerns, we evaluate exchange rate responses using specifications that control explicitly for contemporaneous external shocks and examine multiple exchange rate measures to distinguish domestic transmission from global cross-rate movements.

We first estimate the response of the nominal effective exchange rate (NEER) to identified monetary policy shocks while conditioning on a broader set of external factors, including EU growth, global risk (VIX), global commodity prices, and changes in sovereign spreads, as well as lagged interest rate differentials relative to the ECB (Figure 4.6). Incorporating these controls helps align same-quarter external shocks with the regression and reduces the risk that risk-premium surprises are misattributed to the monetary policy shock. Under this specification, a one percentage point contractionary monetary policy shock leads to a short-lived, statistically significant appreciation of the NEER after two quarters. The response is stronger in the post-2018 subsample, consistent with the increased scope for exchange rate adjustment under wider exchange rate bands.

Figure 4.6 Exchange Rate Channel: NEER (Full sample, Pre- and Post-2018)



Notes: Impulse responses are shown for a contractionary one percentage point monetary policy shock. Shaded areas denote 90 percent confidence bands. An increase in response indicates an appreciation. The specification includes broader external controls relative to the baseline, including EU growth, global risk (VIX), global commodity prices, and changes in sovereign spreads, as well as lagged interest rate differentials relative to the ECB.

To shed further light on the exchange rate response, we examine responses of bilateral exchange rates (Figure 4.7). EUR/MAD, which is the dominant currency in Morocco's exchange-rate reference basket, moves in the same direction as the NEER, with larger short-run elasticities in the post-2018 period, though estimates are less precise in the shorter subsample. In contrast, USD/MAD responses are generally insignificant, consistent with the fact that USD/MAD is largely driven by global USD/EUR movements, making it less informative for assessing Moroccan monetary transmission than NEER and EUR/MAD.

The exchange rate can respond to monetary policy even when UIP is not binding and cross-border arbitrage is limited. With capital controls and a band-based exchange rate regime, interest parity driven persistence is likely to be weak, implying exchange rate responses that materialize with a delay, peak after a few quarters and do not persist, resulting in hump-shaped and relatively short-lived dynamics. Such response patterns are more common in emerging markets with managed exchange rate arrangements—including pegs, stabilization arrangements or horizontal bands—where policy-driven exchange rate movements are partially absorbed over time rather than amplified through arbitrage (for example Checo et al., 2024; Bolhuis et al., 2024).[16]

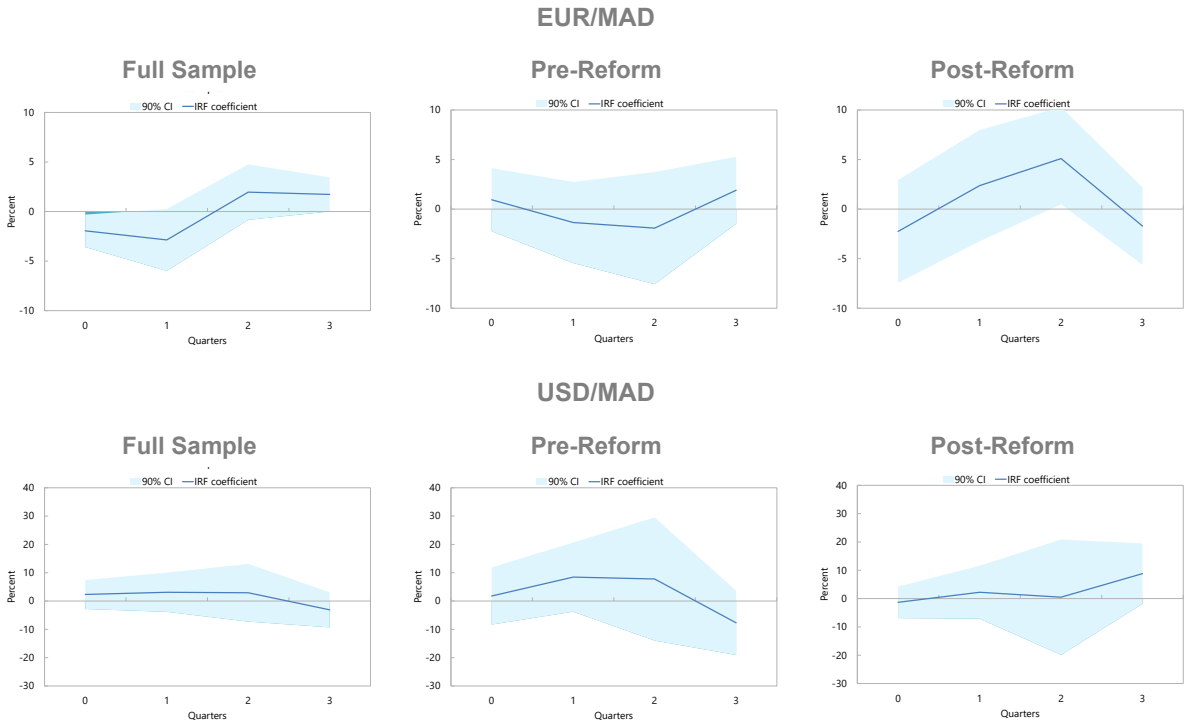
In this context, exchange rate adjustment can operate through domestic mechanisms, including (i) domestic portfolio rebalancing toward dirham denominated assets when domestic yields rise, (ii) tighter liquidity and credit conditions that reduce import related FX demand, (iii) changes in exchange rate expectations—particularly a reduction in perceived deprecation risk or greater confidence in band stability—which lower hedging demand and FX risk premia, and (iv) banks' FX risk management and balance sheet rebalancing in response to monetary policy actions, liquidity conditions and central bank signaling. For instance, consistent with a domestic portfolio channel, the share of FX deposits declines following a contractionary shock after adjusting for mechanical valuation effects (Annex II, Figure A.1).[17]

[16] Checo et al. (2024) document a modest and temporary appreciation following monetary policy tightening in a sample of 18 emerging market economies using survey-based policy-rate forecast surprises. Bolhuis et al. (2024) find that a 100-bps monetary policy shock leads to a sharp and persistent NEER appreciation in advanced economies, but only modest and short-lived exchange rate responses in emerging markets; in several cases the response displays a counterintuitive sign, consistent with the exchange-rate puzzle.

[17] Anecdotal evidence from market participants suggests that additional domestic mechanisms may reinforce the exchange-rate response to monetary policy signals even under capital controls. Banks report that monetary policy tightening and associated domestic rate increases affect clients' currency choices for deposits, borrowing, and hedging, with importers, exporters, and commodity-exposed firms adjusting the currency composition and maturity of their balance sheets in response to exchange-rate expectations within the fluctuation band. Banks also indicate more active management of their net open foreign exchange…

These mechanisms provide an interpretation of the estimated exchange rate responses that do not rely on strong UIP arbitrage. Overall, the NEER and the bilateral rates responses indicate that the exchange rate channel is more apparent post-2018, and is likely to operate through domestic financial conditions rather than through UIP-driven arbitrage. This is reflected in moderate exchange rate effects that tend to dissipate over short horizons, consistent with Morocco's band-based exchange rate framework.

Figure 4.7. Exchange Rate Channel: Bilateral Rates (Full sample, Pre- and Post-2018)



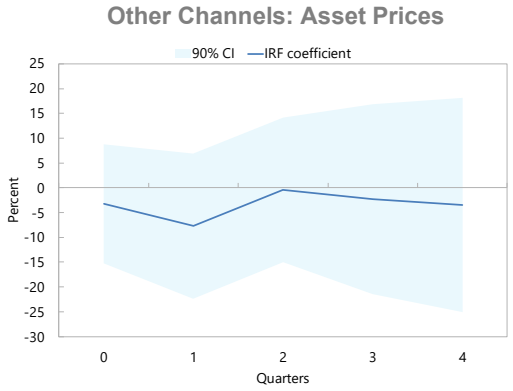
Notes: Impulse responses are shown for a contractionary one percentage point monetary policy shock. Shaded areas denote 90 percent confidence bands. An increase in response indicates an appreciation. The specification includes broader external controls relative to the baseline, including EU growth, global risk (VIX), global commodity prices, and changes in sovereign spreads, as well as lagged interest rate differentials relative to the ECB.

Asset Price Channel

We find limited evidence that monetary policy shocks transmit through equity prices. Stock market indices show no statistically significant response to a contractionary monetary policy shock, with point estimates small relative to uncertainty (Figure 4.8). This muted response is consistent with the stylized facts documented in Section 3: despite the Casablanca Stock Exchange's relatively sizable capitalization, market liquidity is limited, household participation is low (due to limited financial literacy), the number of listed firms is relatively small, trading is concentrated in a few sectors, and institutional investor activity tends to be relatively passive. These structural features constrain the scope for wealth and equity valuation effects to meaningfully amplify monetary policy transmission through equity prices.

…positions—within regulatory and internal limits—through higher-frequency rebalancing and two-way positioning following the widening of the exchange-rate band, reflecting sensitivity to domestic liquidity conditions and monetary-policy signals.

Figure 4.8 Asset Price Channel: Stock Market

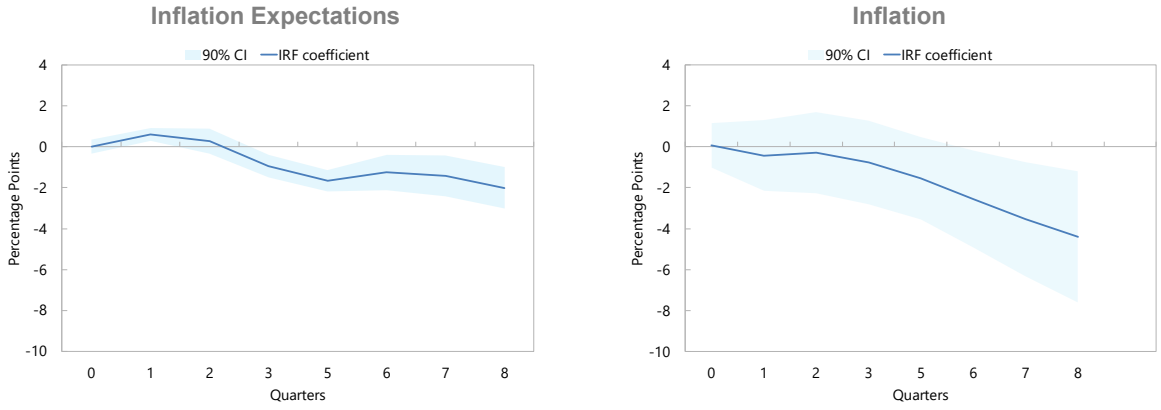


Notes: Impulse responses are shown for a contractionary one percentage point monetary policy shock. Shaded areas denote 90 percent confidence bands. Full sample results are shown.

Expectations Channel

Inflation expectations decline following a contractionary monetary policy shock, indicating some transmission through the expectations channel, though the response is gradual and becomes statistically significant only at medium horizons (Figure 4.9). Expectations are measured as 8-quarter-ahead inflation expectations of financial sector experts. The timing of the decline suggests that expectations are at least partly reactive—adjusting as realized inflation falls—rather than strongly forward-looking. This pattern is consistent with the stylized facts in Section 3, which point to broadly anchored inflation expectations in a low-inflation environment, supported by the exchange rate anchor, with expectations moving more noticeably during periods of sizable inflationary pressures.

Figure 4.9 Expectations Channel: Inflation Expectations



Notes: Impulse responses are shown for a contractionary one percentage point monetary policy shock. Shaded areas denote 90 percent confidence bands. Inflation expectations are measured as 8-quarter-ahead expectations from Bank Al-Maghrib's financial-sector experts survey.

4.6. Robustness Checks

We assess the robustness of our results to alternative identification choices, specification assumptions, and sample definitions. Across these checks, the qualitative conclusions remain unchanged: monetary policy shocks reduce output and inflation, with inflation responding more slowly than output; transmission is stronger post 2018; the interest rate channel is the most active; the credit channel is heterogeneous; and the exchange rate responses remain present but moderate under a band-based exchange rate arrangement.[18]

Alternative monetary policy measure. As a validation exercise, we re-estimate the baseline local projection model using changes in the actual policy rate instead of the identified monetary policy shocks. While inflation responses remain broadly similar—though somewhat smaller in magnitude—output responses become counterintuitive, with economic activity increasing following policy rate hikes. This result is consistent with the endogeneity concerns that motivate the identification strategy, as observed policy rate changes reflect both the systematic response of monetary policy to evolving macroeconomic conditions and unexpected policy actions. By contrast, the identified shock measure isolates deviations from the systematic component of policy, yielding responses that are more consistent with standard monetary transmission mechanisms and the broader empirical literature. These findings provide additional support for the use of identified monetary policy shocks in the Moroccan context.

Regime-shift interaction specification. As a robustness check, we re-estimate the local projections over the full sample using an interaction term between the monetary policy shock and a post-2018 reform indicator. This specification preserves the full sample and provides a formal test of whether transmission differs across periods. We treat this approach as complementary rather than baseline because it summarizes a multi-year reform process—including greater exchange rate flexibility, enhanced communication and transparency, operational improvements, and a strengthened focus on price stability—through a single regime indicator. The results point to stronger output transmission in the post-reform period, while differences in inflation responses are smaller and less precisely estimated. These findings are consistent with the view that reforms strengthened the transmission mechanism, particularly through channels affecting real economic activity. The more muted results for inflation may reflect the gradual nature of price adjustment and the continued importance of food and administered prices in Morocco's inflation process. Overall, the interaction specification points to stronger post-reform transmission, although the estimated differences are characterized by greater uncertainty than in the baseline specification.

Alternative shock specifications. First, we estimate monetary policy shocks using alternative reaction function specifications that augment the baseline information set with additional external and domestic variables relevant for a small open economy, including the NEER and bilateral exchange rates, the ECB policy rate, and international reserves. The resulting shock series are highly correlated with the baseline measure, and the estimated impulse responses for output, prices, and key financial indicators are qualitatively similar (Annex II Table A2, Figure 4.1b).

Controls for external conditions. Second, we verify that results are not driven by omitted external shocks by expanding the control set in the LP regressions to include contemporaneous global risk and external demand indicators (including VIX and EU growth), global commodity prices, and changes in sovereign spreads, in

[18] The additional results are not reported for brevity but are available from the authors upon request.

addition to lagged foreign interest rate differentials. This check is particularly important for exchange-rate responses, where global risk and country-risk surprises can otherwise be confounded with domestic monetary shocks. Under the expanded controls, the NEER continues to appreciate following a contractionary shock, with an effect that remains statistically significant but modestly smaller and short-lived (Figure 4.6).

Alternative outcome measures. Third, we confirm that the estimated macroeconomic effects are robust to reasonable alternatives for real activity and inflation, including the use of non-agricultural output measures and alternative inflation series. Results remain consistent in sign, timing, and broad magnitude.

Lag structure and sample sensitivity. Fourth, we assess sensitivity to dynamic specification choices by varying the lag length in the LP regressions. We also test the stability of results to the inclusion of a COVID-period control and to excluding the most disrupted quarters. These exercises yield similar impulse response patterns, with wider confidence bands in shorter subsamples as expected.

Small-sample inference and standard errors. As is standard in local-projections settings, we construct confidence intervals using Newey–West heteroskedasticity- and autocorrelation-consistent standard errors to account for overlapping dependent variables and serial correlation across horizons. Given the relatively short effective sample—particularly in post-2018 subsamples—confidence bands widen at longer horizons, reflecting increased estimation uncertainty rather than instability of the underlying point estimates. Importantly, across specifications, subsamples, and robustness checks, the sign, timing, and relative magnitude of the impulse responses remain stable, supporting the robustness of the main conclusions.

5. Conclusion

This paper assesses Morocco's monetary policy transmission in the context of recent modernization of the policy framework and a gradual transition toward inflation targeting. Using a two-step empirical strategy—forecast-based identification of monetary policy shocks and local projections to trace dynamic effects—we document that transmission to output and inflation has strengthened in the post-2018 period relative to earlier years, though responses remain delayed for output and comparatively muted for inflation.

The channel analysis helps interpret these aggregate patterns. The interest rate channel is the most active, with policy shocks transmitting clearly to short term money market rates, government securities, and bank lending rates, consistent with improved operational control at the short end of the yield curve. The credit channel is partially effective and heterogeneous: responses are more immediate for short-term and services-related lending, while longer-term and program-linked segments adjust more gradually, consistent with differences in maturity structure, repricing frequency, and loan characteristics. The exchange rate channel is more apparent post 2018; exchange rate responses remain moderate and short lived, consistent with a band-based arrangement and imperfect capital mobility, and appear to operate primarily through domestic financial conditions rather than UIP driven arbitrage. In contrast, the asset price channel remains weak, and the expectations channel shows only modest and gradual adjustment, consistent with the stylized evidence of anchored but partly reactive expectations in a low inflation environment.

These findings carry several policy implications for Morocco's continued transition toward inflation targeting.[19] First, the strengthened control of short-term rates provides a solid operational foundation for transmission, but enhancing pass through—particularly to deposit rates and longer-term lending conditions—would help amplify real effects. Second, structural features of inflation—especially the large weight of food and administered prices—imply that headline inflation is likely to respond gradually to monetary policy, underscoring the importance of clear communication and expectation management. Third, further deepening of financial markets and continued development of FX market infrastructure can broaden the set of channels through which policy affects financing conditions and improves the effectiveness of the exchange rate channel under greater flexibility.

The analysis has limitations. The post-2018 sample is relatively short, and several disaggregated financial and price series are only available for more recent years, which limits precision and constrains the analysis of nonlinear transmission and state dependence (including tightening versus easing episodes or high- versus low-inflation regimes). In particular, the limited number of distinct monetary policy cycles reduces the scope for credibly estimating asymmetric effects or richer dynamic responses. As additional post-reform observations accumulate, future work could explore instrumental-variables local projections (LP-IV) to further address endogeneity concerns, though the feasibility and gains from such approaches depend critically on sample length and instrument strength. A complementary and potentially fruitful extension would be to exploit disaggregated CPI data to assess transmission to monetary-relevant price components, such as market-determined non-food services and unregulated goods, which may respond more visibly to policy shocks than headline inflation or available core measures that still embed food and administered prices. These extensions would help further clarify the mechanisms behind the gradual and muted inflation responses documented in this paper, conditional on data availability and sample length.

[19] Related policy considerations were discussed in the 2026 Article IV Consultation Staff Report (see paragraph 22 and Box 2).

Annex I. Data Sources

Table A1. Data Sources and Description

Variables	Description	Source
Headline inflation forecast	4-quarter-ahead forecast of headline inflation.	Bank Al-Maghrib
Core inflation forecast	4-quarter-ahead forecast of core inflation.	Bank Al-Maghrib
Growth forecast	4-quarter-ahead forecast of real GDP growth.	Bank Al-Maghrib
Nonagricultural GDP forecast	4-quarter-ahead forecast of non-agricultural GDP growth.	Bank Al-Maghrib
Policy rate	Bank Al-Maghrib's policy interest rate.	Bank Al-Maghrib
CBOE Market Volatility Index (VIX)	The Chicago Board of Exchange's S&P500 implicit volatility index.	Bloomberg
Global Commodity Price Index	Goldman Sachs global commodity price index.	Bloomberg
Precipitation Anomaly	Country-level precipitation anomaly used as a control variable.	Chen et al. (2002)
Temperature Anomaly	Country-level temperature anomaly used as a control variable.	GISTEMP (NASA)
COVID-19 Dummy	Dummy variable which takes value 1 (0) for time period during (not during) the COVID-19 pandemic.	Authors' calculations
Euro Area Policy Rate	ECB Main refinancing rate	Haver, European Central Bank
Euro Area Growth	Growth rate of euro area (EA20) real GDP.	Haver, Eurostat
Sovereign Spread	Weighted-average sovereign spread (basis points) across MAD- and USD-denominated bonds.	IMF Sovereign Spread Monitor
NEER	Nominal effective exchange rate.	Haver, IMF IFS
REER	Real effective exchange rate, adjusted by relative consumer prices.	Haver, IMF IFS
Required Reserves	Required reserve ratio, average percent.	Haver, Bank Al-Maghrib
USD/MAD	Dirham–U.S. dollar bilateral exchange rate (average).	Haver, Bank Al-Maghrib
EUR/MAD	Dirham–euro bilateral exchange rate (average).	Haver, Bank Al-Maghrib
International Reserves	Gross international reserves (level).	Haver, Bank Al-Maghrib
Bank Lending Rates	Lending rates by borrower type and loan purpose.	Haver, Bank Al-Maghrib
Bank Credit	Bank loans by sector, maturity, purpose, and institutional sector.	Haver, Bank Al-Maghrib
Government Securities	Treasury yields at 13-week, 26-week, 1-year, 2-year, 5-year, and 10-year maturities.	Haver, Bank Al-Maghrib
Bank Deposit Rates	Deposit rates on 6- and 12-month accounts and cash certificates	Bank Al-Maghrib
Inflation Expectations	8-quarter-ahead inflation expectations of financial sector experts, percent.	Haver, Bank Al-Maghrib
Stock Market Index	Casablanca All Share Index (EOP, Dec-31-91=1000)	Haver, Bank Al-Maghrib
FX Deposits	Foreign currency deposits as a share of total bank deposits, EOP, NSA.	Haver, Bank Al-Maghrib

Annex II. Additional Results

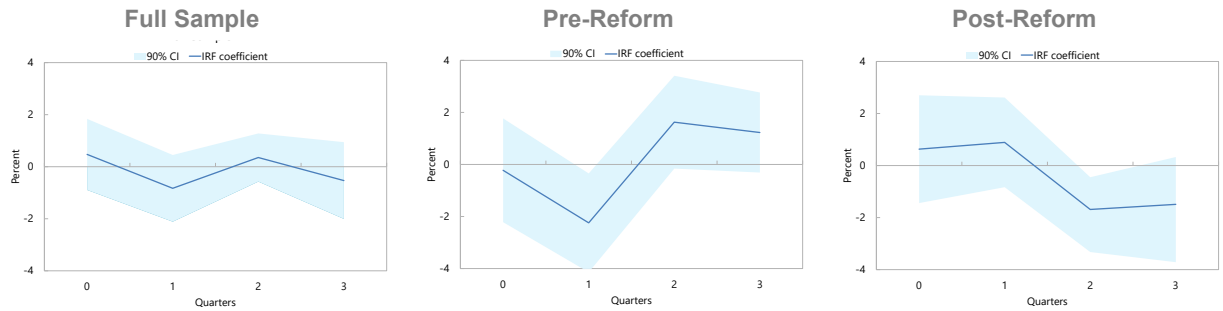
Table A2. Baseline and Alternative Model Specifications for Monetary Policy Shocks

Variables	Baseline	Model 1	Model 2	Model 4	Model 5	Model 6	Model 7
Inflation forecast	0.0415**	0.0390**	0.0418**	0.0434**	0.0285*	0.0413**	0.0414**
	-0.0183	*-0.0191*	*-0.0181*	*-0.0193*	*-0.0154*	*-0.0186*	*-0.0185*
Growth forecast	0.0470***	0.0475***	0.0477***	0.0315*	0.0561***	0.0442***	0.0471***
	-0.0135	*-0.0137*	*-0.0134*	*-0.0171*	*-0.0113*	*-0.0137*	*-0.0136*
Δ log CPI (lagged once)	0.0267	0.0269	0.0227	0.0213	0.0236	0.02	0.0266
	-0.0269	*-0.0273*	*-0.0268*	*-0.0273*	*-0.0223*	*-0.0274*	*-0.0271*
Δ log CPI (lagged twice)	-0.0119	-0.0107	-0.00508	-0.0158	-0.00692	-0.00894	-0.0119
	-0.0252	*-0.0256*	*-0.0252*	*-0.0263*	*-0.0209*	*-0.0254*	*-0.0254*
Δ log GDP (lagged once)	-0.0196**	-0.0194**	-0.0164*	-0.0149	-0.0175**	-0.0170*	-0.0198**
	-0.0091	*-0.0093*	*-0.00923*	*-0.00964*	*-0.00756*	*-0.00937*	*-0.00958*
Δ log GDP (lagged twice)	-0.0151*	-0.0141*	-0.0140*	-0.0101	-0.0139**	-0.0149*	-0.0151*
	-0.00784	*-0.00816*	*-0.0079*	*-0.00866*	*-0.00651*	*-0.00788*	*-0.0079*
Δ log ER_USD (lagged once)		-0.000518					
		-0.005					
Δ log ER_USD (lagged twice)		-0.00262					
		-0.00526					
Interest rate (lagged once)	0.252*	0.257*	0.2	0.262*	0.0657	0.250*	0.253*
	-0.134	*-0.137*	*-0.138*	*-0.134*	*-0.116*	*-0.135*	*-0.136*
Interest rate (lagged twice)	-0.339**	-0.344**	-0.296**	-0.335**	-0.12	-0.339**	-0.340**
	-0.134	*-0.136*	*-0.136*	*-0.134*	*-0.118*	*-0.135*	*-0.135*
Δ inflation forecast				0.00438			
				-0.0281			
Δ growth forecast				0.0219			
				-0.0149			
Δ log NEER (lagged once)			-0.0267*				
			-0.0148				
Δ log NEER (lagged twice)			-0.0019				
			-0.0153				
Δ ECB policy rate					0.213***		
					-0.0392		
Δ International reserves (lagged once)						0.00056	
						-0.00212	
Δ International reserves (lagged twice)						-0.00303	
						-0.00222	
Δ Reserves Requirements							0.00139
							-0.0229
Constant	-0.00516	-0.00467	0.0196	0.00252	-0.101	0.0103	-0.00564
	-0.0736	*-0.0746*	*-0.0759*	*-0.0759*	*-0.0636*	*-0.0759*	*-0.0746*
Observations	72	72	72	72	72	72	72
R-squared	0.454	0.458	0.483	0.473	0.63	0.471	0.455

Standard errors in italics

*** p<0.01, ** p<0.05, * p<0.1

Figure A1. FX Deposits Response to MP Shocks (Domestic portfolio shifts)



Notes: Impulse responses are shown for a contractionary one percentage point monetary policy shock. Shaded areas denote 90 percent confidence bands. The dependent variable is the share of FX deposits in total deposits. To avoid mechanical revaluation effects from exchange rate movements, the series is valuation-adjusted using lagged Δ log(EUR/MAD).

Annex III. Interpreting Local Projection Estimates in Small Samples

This annex summarizes insights from recent literature on the behavior of local projections (LP) estimates in small samples. A growing body of work shows that, when the dependent variable is persistent and the sample size is limited, least-squares LP impulse responses may be subject to attenuation bias, particularly at longer horizons. Importantly, this bias tends to be downward, implying that estimated responses are smaller in magnitude than the true underlying effects. Herbst and Johannsen (2024) document that small-sample bias in LPs does not alter the qualitative shape or timing of impulse responses, but primarily attenuates their magnitude. Related discussions in Jordà and Taylor (2025) emphasize that LPs remain robust to model misspecification and informative about dynamic patterns even in finite samples, with small-sample issues primarily affecting precision rather than qualitative inference.

This consideration is relevant for the interpretation of our results. Given Morocco's limited sample size and the persistence of key macroeconomic variables, the estimated responses to monetary policy shocks—particularly for output and inflation at medium to longer horizons—are likely to be conservative. The qualitative patterns highlighted in the main text, including the direction, timing, and relative strength of responses across channels, remain informative, while the true effects may be somewhat larger than implied by point estimates.

References

Bank Al-Maghrib (2019): Statute of Bank Al Maghrib. Available at: https://www.bkam.ma/en/content/download/668407/8534171/Statut%20BAM%20A5%20ANG.pdf

Bank for International Settlements (2008): Transmission Mechanisms for Monetary Policy in Emerging Market Economies, BIS Papers No. 35. https://www.bis.org/publ/bppdf/bispap35.pdf

Bank for International Settlements (2024): 20 Years of Central Bank Communications, and Lessons for the Future, BIS Papers No. 143. https://www.bis.org/publ/bppdf/bispap143_e.pdf

Bems, R., Caselli, F., Grigoli, F., and Gruss, B. (2020): Gains from Anchoring Inflation Expectations: Evidence from the Taper Tantrum Shock, Economics Letters, 188, 108820.

Bernanke, B. S., and Blinder, A. S. (1992): The Federal Funds Rate and the Channels of Monetary Transmission, American Economic Review, 82(4), 901–921.

Bernanke, B. S., Boivin, J., and Eliasz, P. (2005): Measuring the Effects of Monetary Policy: A Factor Augmented Vector Autoregressive (FAVAR) Approach, The Quarterly Journal of Economics, 120(1), 387–422. https://doi.org/10.1162/0033553053327452

Bolhuis, M., Das, S., and Yao, J. (2024): Monetary Policy Transmission in Emerging Markets, IMF Working Paper, Washington, DC: International Monetary Fund.

Brandão Marques, L., Gelos, G., Harjes, T., Sahay, R., and Xue, Y. (2020): Monetary Policy Transmission in Emerging Markets and Developing Economies, IMF Working Paper 035/2020, Washington, DC: International Monetary Fund.

Brito, S., Carrière-Swallow, Y., and Gruss, B. (2018): Disagreement about Future Inflation: Understanding the Benefits of Inflation Targeting and Transparency, IMF Working Paper WP/18/24. https://doi.org/10.5089/9781484339718.001

Checo, A., Grigoli, F., and Sandri, D. (2024): Monetary Policy Transmission in Emerging Markets: Proverbial Concerns, Novel Evidence, BIS Working Paper (March 2024).

Christiano, L. J., Eichenbaum, M., and Evans, C. L. (1996): The Effects of Monetary Policy Shocks: Evidence from the Flow of Funds, Review of Economic Statistics, 78(1), 16–34.

Cochrane, J. H., and Piazzesi, M. (2002): The Fed and Interest Rates: A High Frequency Identification, American Economic Review, 92(2), 90–95.

Deb, P., Estefania Flores, J., Firat, M., Furceri, D., and Kothari, S. (2023): Monetary Policy Transmission Heterogeneity: Cross Country Evidence, IMF Working Paper 23/204, Washington, DC: International Monetary Fund.

Eklou, K. (2023): The Anatomy of Monetary Policy Transmission in an Emerging Market, IMF Working Paper 2023/146, Washington, DC: International Monetary Fund.

European Central Bank (2021): Clear, Consistent and Engaging: ECB Monetary Policy Communication in a Changing World, ECB Occasional Paper No. 274. https://www.ecb.europa.eu/pub/pdf/scpops/ecb.op274~9aca14e6f6.en.pdf

Gertler, M., and Karadi, P. (2015): Monetary Policy Surprises, Credit Costs, and Economic Activity, American Economic Journal: Macroeconomics, 7(1), 44–76.

Herbst, E., and Johannsen, B. (2024): Bias in Local Projections, Federal Reserve Bank Working Paper.

International Monetary Fund (2018): World Economic Outlook: Challenges to Steady Growth, October 2018. See Chapter 3: Challenges for Monetary Policy in Emerging Markets as Global Financial Conditions Normalize.

International Monetary Fund (2022): Morocco: Central Bank Transparency Code Review, IMF Country Report No. 22/380, Washington, DC.

International Monetary Fund (2023): Monetary Policy: Where Does the Middle East and Central Asia Stand? In Regional Economic Outlook: Middle East and Central Asia, April 2023, pp. 21–36.

Jordà, Ò. (2005): Estimation and Inference of Impulse Responses by Local Projections, American Economic Review, 95(1), 161–182.

Jordà, Ò., and Taylor, A. M. (2025): Local Projections, Journal of Economic Literature, 63(1), 59–110. https://doi.org/10.1257/jel.20241521

Kuttner, K. N. (2001): Monetary Policy Surprises and Interest Rates: Evidence from the Fed Funds Futures Market, Journal of Monetary Economics, 47(3), 523–544.

Leigh, D., and Xu, R. (2025): Monetary Policy Transmission to Lending Rates: Evidence from Brazil, IMF Working Paper 2025/152, Washington, DC: International Monetary Fund.

Li, W. (2026): Beyond Binary: A Policy Intensity Measure of Capital Flow Management, IMF Working Paper 26/21, Washington, DC: International Monetary Fund.

Mishkin, F. S., and Schmidt Hebbel, K. (2002): Inflation Targeting in Practice: Strategic and Operational Issues and Application to Emerging Market Countries, IMF Working Paper WP/02/102.

Roger, S., and Stone, M. (2005): On Target? The International Experience with Achieving Inflation Targets, IMF Working Paper WP/05/163.

Romer, C. D., and Romer, D. H. (2004): A New Measure of Monetary Shocks: Derivation and Implications, American Economic Review, 94(4), 1055–1084.

Sims, C. A. (1980): Macroeconomics and Reality, Econometrica, 48(1), 1–48.

Stock, J. H., and Watson, M. W. (2018): Identification and Estimation of Dynamic Causal Effects in Macroeconomics Using External Instruments, Economic Journal, 128(610), 917–948.

Zelmer, M., Schaechter, A., Stone, M. R., and Carare, A. (2002): Establishing Initial Conditions in Support of Inflation Targeting, IMF Working Paper WP/02/102, Washington, DC: International Monetary Fund.



PUBLICATIONS

Effectiveness of Monetary Policy Transmission in Morocco amid Reforms to the Monetary Policy Framework

Working Paper No. WP/2026/223